SoK: Transfer Boundaries of LLM and Agent Side-Channel Defenses

***Abstract*—Side-channel defenses for LLM serving and agent systems are evaluated at one operating point and then reused at another: a different emitter, tool dispatcher, model or observer. We ask which changes invalidate a measured defense result. Our unit of analysis is the evaluated result record, which binds an outcome to its target, receiver, intervention boundary, attacker fit, workload and cost object. A map of 82 study families, 39 source-bound records and a nine-item reuse audit show that the evidence deployment needs is the evidence least often reported: inference is bound for 17 of 20 mitigation families, but matched adaptation for only 2 of 11 applicable families and utility at the protected operating point for 3 of 17; hardware and temporal transfer remain unresolved for most families. Four controlled experiments then exercise each boundary: upstream padding does not protect a re-emitting frontend, idempotent cover does not preserve state, timing exposes cover that length-only evaluation scores at chance, and defense calibration does not transfer as a scalar. Moving a calibrated defense from MiniMax to GLM raises cancellation balanced accuracy from 50.0% to 70.8% while lowering return from 67.3% to 50.0%: protection is exchanged between targets rather than uniformly degraded. We distill these results into a decision procedure: identify the boundary a deployment changes, require evidence bound on that boundary, and otherwise re-measure it.**

1. Introduction

Encryption conceals message content, yet LLM engines and agents expose packet sizes, cache timing, visited domains and tool sequences [1], [2], [3], [4], [5], [6]. Defenses follow: padding, batching, cache isolation, decoy queries and action humanization. Each is evaluated at one receiver, execution boundary and workload, yet its headline number is cited as a property of the mechanism. A padding result measured on an upstream link says nothing about a frontend that re-emits decoded text; decoys whose read responses are discarded say nothing about cover that executes writes; and the overhead of a semantic cache does not price prefix isolation reported in the same paper.

We therefore systematize defense evidence at the level of the *evaluated result record* rather than the paper. A record binds one outcome to its target, receiver projection, intervention boundary, attacker fit, workload and cost object (§3). Three instruments operate on these records: a map of 82 study families by signal origin, receiving view and threat polarity; a nine-item audit of what each mitigation family binds; and a six-axis matrix of reported holdouts. Together they answer one question: *which changes invalidate a measured defense result?*

Contributions.

- **Record-level systematization (§§2–7).** Separating records exposes within-paper substitution: 6 of 13 same-family pairs in the original 33-record audit measure different primitives, targets or cost objects, and only 16 of 102 related record pairs support even a descriptive comparison. Reuse evidence is scarcest where deployment needs it: matched adaptation is bound for 2/11 applicable mitigation families and protected-point utility for 3/17, while hardware and temporal transfer remain unresolved for most families.
- **Four boundary demonstrations (§8).** Controlled experiments on a retail agent workload instantiate the emitter, executed-state, calibration and attacker-observation boundaries identified by the audit, each priced with its own cost object.
- **Directional calibration transfer.** Moving calibration from MiniMax to GLM changes cancellation/return BA from 50.0/67.3% to 70.8/50.0%: protection moves in opposite directions, so no single transfer score summarizes it. Write-local shaping is inexpensive (1.45–$1.51\times$ bytes) but leaves return BA at 61.3–61.6%, and a fresh live-planner study (37/50 versus 33/50 successes) bounds the utility claim of prefix cover.

Adaptive attackers, distribution shift and online cost evaluation are established website-fingerprinting (WF) practice [7], [8]; we apply them to records rather than claim them. Exps 1–2 deliberately reproduce known mechanisms so that their cost and state consequences can be measured; Exps 3–4 measure agent-specific operating points. Table 1 links each research question to its finding.

Scope of claims. The real-agent target is a requested operation category in one retail environment with a known 53-customer cohort. Measurements establish fitted-suite inference, logical state conformance and scoped resource costs; a separate 100-episode live study measures prefix-cover utility under injected deadlines. Deployment-wide security and full joint utility require the validation specified in §9.3.

Figure 1 places trace and action paths on the same agent–tool chain, making the receiving representation and the execution boundary explicit before a defense is selected.

TABLE 1. RESEARCH QUESTIONS, SYSTEMATIZATION FINDINGS AND WHERE EACH IS ESTABLISHED. EXPERIMENTS TEST THE COMPARISON RULES; THE SOK CONTRIBUTION IS THE RECORD-LEVEL SYNTHESIS.

Question	Finding	Evidence
RQ1: targets and observables	A common signal can serve confidentiality, attribution, evasion or authorized diagnostics; polarity changes the protected interest. Signal generation clusters at serving (32 families) and orchestration (19), whereas observation clusters at application, hardware and transport.	Complete 58-family core map (§4, Table 6).
RQ2: origin versus access	Packet-size cover does not bind call order; plaintext re-emission creates a fresh size trace; shared call projections do not establish unchanged state. In the original pairwise audit, 6 of 13 same-family pairs measure different primitives, targets or cost objects.	39-record ledger (§5); NetEcho/SplitDecoy and WRA/AHB lenses (§7); Exps 1–2.
RQ3: portability	The archive resolves task/input holdouts in 13 of 82 families and model holdouts in three; 73 hardware and 71 temporal entries remain unresolved. Input separation does not certify a changed generator.	Six-axis matrix (§6, Appendix D); calibration-transfer matrix in Exp 3.
RQ4: protection, state and cost	Inference is bound far more often than reuse: matched adaptation 2/11 applicable families, utility at the protected operating point 3/17, cost 11/19.	Nine-item, 20-family audit (§5); Exps 3–4 leakage/state/cost contrasts.

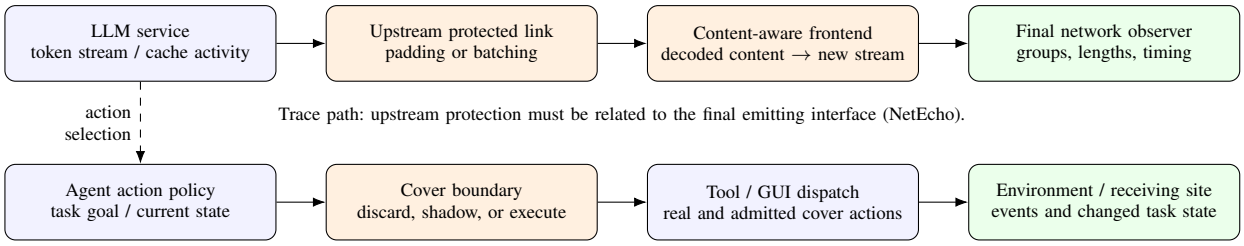


Figure 1. Observation and execution boundaries. The four demonstrations test emitter placement, stateful execution/link composition, real-agent operation signatures, and timing with matched transport controls.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions: which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems (RQ1); how signal generation origin relates to receiver observation permissions (RQ2); which holdouts establish portability across tasks, configurations and hardware (RQ3); and which evidence binds workflow protection, logical state preservation and resource cost, distinguishing these from full end-to-end planner utility (RQ4).

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [9], [10], [11], [12], [13]. Table 2 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

(II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [14], [15]. These roles are mapped per core family, alongside generating and receiving layers, in Table 6.

2.2. Corpus construction and reconciliation

Forty seeds, venue-targeted proceedings screening and one-hop citation snowballing produced an initial map of 60 families (43 core, 17 adjacent). A supplementary database and bulk keyword search returned 615 occurrences and 565 unique records; auditing its 204 exclusions rerouted four records, and the search added 15 core and seven adjacent families. The final collection has 58 core and 24 adjacent families.

2.3. Source-bound result records

Each result record binds its target, observation boundary, operating point and source section/table/figure/paragraph locator. The authors checked every record and audit entry against these sources. Appendix B indexes all 39 records with their measurement modes; Appendix A describes the supporting artifact.

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

3. Result Records and Comparability Rules

The atomic unit of our systematization is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0: clean-condition fit; D_2: protected-condition fit), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A study family contributes multiple records whenever it evaluates distinct execution modes (e.g., stored capture vs. live loopback in Dangol [16]), distinct mechanisms (e.g., K-prefix KV simulation vs. GPTCache semantic de-identification in Early Bird [3]), or separately scoped performance benchmarks (e.g., SafeKV [17] and OTRO [18]).

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (U), unavailable (A) or unverified (?). Table 5 illustrates a complete record-to-judgment lineage.

From contracts to an audit. Section 5 operationalizes these contracts as nine reuse items with a shared evidence-status alphabet (Table 8) and applies them to every mitigation family (Table 9).

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

4. Corpus Map Across 82 Study Families

4.1. Generating layers, receiving projections and threat polarities

Table 6 maps all 58 core families by signal-generating layers, receiving projections and threat polarity. Multi-layer families retain every reported layer in their generating and observing sets.

Cross-layer synthesis. A remote network observer, a same-site script and a privileged diagnostic hook expose

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [16].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about $13\times$ bytes and $11\times$ latency in that condition.
Unresolved fields	Objective terminal task utility U; precise paired live-refit comparison ?.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

different views even when they share an attribution target. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate why receiving permissions and protected objectives must accompany the generating layer [6], [14], [34].

Layer incidence highlights where the systematization's comparison problem arises. In the 58-family map, signal-generation labels cluster at P2 serving/runtime (32 families) and P5 orchestration (19); receiving views cluster at P4 application/API (18), P1 hardware (16), and P3 transport (14). Generation at P3/P4 is rare (one family each), despite frequent observation at those layers. This source/receiver asymmetry makes the transfer question concrete: protection at a frequent source layer cannot be presumed to cover a different receiving projection. The map records layer sets, so it does not assign a causal edge to each source/receiver pair.

Purpose labels further show why a single "agent side channel" category is insufficient: Type I appears 41 times, Type II 13, Type III 2, and Type IV 5. The 32 map-only families extend coverage beyond mitigation comparisons: 15 list serving origins, 11 hardware and 11 orchestration; 19 concern confidentiality, nine attribution, one evasion and all five authorized-diagnostic families. Mapping these families locates privacy targets, operator capabilities and candidate validation environments even when no scoped mitigation record enters the ledger. Counts are overlapping family-label incidences; the audit then applies its reuse rules to the 39 scoped result records.

4.2. Relationship to prior systematizations

Prior surveys map compound-system threats, TEE boundaries and privacy contracts [66], [67], [68]. WF already supplies feature leakage analysis, cost trade-offs and environment-shift tests [69], [7], [70], [71]. Mathews et al. re-evaluate nine defenses with stronger attacks and an on-network DynaFlow implementation [8]. Adaptive attacks and online-cost evaluation are established foundations.

What this synthesis adds. Our unit is a *scoped result*, so execution mode, protected state and cost remain attached when one paper reports several mechanisms. The 39-record census exposes these within-paper substitutions; the 82-family holdout matrix separates evidence for each transfer axis. The empirical additions are target-dependent calibration transfer, the cost/residual-inference boundary of write-local shaping, and fresh planner behavior after executed responses are withheld by deadlines. WF packet-shaping costs alone do not measure these tool-state and replanning outcomes.

5. Comparability Audit Across All 58 Core Families

5.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual $+32$ map-only families. The ledger holds 39 records: 33 from the 20 mitigation families and one from each contextual family—SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [57], [64], [65], [23], [36], [44]—which supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than an audited mitigation claim. The 32 map-only families comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound quantitative assertions and SplitDecoy's boundary case. Table 7 groups mechanisms; its eight rows are mechanism groups, not sampled studies.

Whole-ledger synthesis. Each record has one primary measurement mode: 20 executed collections/prototypes, six trace transformations, five simulations/derived estimates, six component/calibration measurements and two unresolved placements (Table 21). Eleven of the 20 mitigation families contribute more than one record, and nine of them span more than one mode; this is why a mechanism name cannot carry utility or cost between records. Executed collection establishes its stated execution scope, not automatically full-workflow utility.

5.2. Within-family and within-cohort record comparisons

The pairwise audit covers the 33 records in the ledger before the final search (27 mitigation, six contextual); the six records that search added contribute two further same-family pairs (`CACHEPRUNE-selective`/`-context` and `OTRO-unpadded`/`-epochpad`). These added pairs separate direct recovery from contextual guessing, and measured unpadded latency from derived padding cost, respectively; the numerical pairwise counts below retain the original 33-record denominator. Of the $\binom{33}{2} = 528$ pairs, 426 cross unrelated mechanism cohorts and are out of scope by

TABLE 6. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

Core family	$G \to O$	Type
hardware generation, P1 (8)		
Edge EM leakage [19]	$1 \to 1$	I
Energon [20]	$1 \to 1$	II
MoEcho [21]	$1 \to 1$	I
I Know What You Said [22]	$1 \to 1$	I
Behind Bars [23]	$1 \to 1$	I
PAGE404 [24]	$1 \to 1$	I
Spill the Beans [25]	$1 \to 1$	I
ArchSnoop [26]	$1 \to 1$	II
hardware and service, P1/P2 (8)		
SparLeak [27]	$1,2 \to 1$	I
Shape and Substance [28]	$1,2 \to 1$	I
TDXRay [29]	$1,2 \to 1$	I
SparSEEty [30]	$1,2 \to 1$	I
OTRO [18]	$1,2 \to 1$	I
Detokenization Leaks [31]	$1,2 \to 1$	I
LLM-WFIN [32]	$1,2 \to 1,3$	I
LLM-serving RCA [33]	$1,2 \to 2$	IV
service generation, P2 (13 of 22)		
SplitDecoy [34]	$2 \to 2$	I
KVGov [35]	$2 \to 2,4$	I
LLMs Have Rhythm [36]	$2 \to 3$	II
Remote timing [37]	$2 \to 3$	I
NetEcho [38]	$2 \to 3$	I
Prompt Echo [39]	$2 \to 3$	I
Speculative decoding [40]	$2 \to 3$	I
Remote keylogging [1]	$2 \to 3$	I
Whisper Leak [41]	$2 \to 3$	I
Cache contention [42]	$2 \to 4$	I
Routing metadata [43]	$2 \to 4$	I
SafeKV [17]	$2 \to 4$	I
ML privacy channels [44]	$2 \to 4$	I
service generation, P2 (9)		
Early Bird [3]	$2 \to 4$	I
Prompt-cache audit [45]	$2 \to 4$	I
InputSnatch [46]	$2 \to 4$	I
Leaky Language Models [47]	$2 \to 4$	I
PromptPeek [2]	$2 \to 4$	I
SpliceLeak [48]	$2 \to 4$	I
Time Will Tell [49]	$2 \to 4$	I
CachePrune [50]	$2 \to 4$	I
MoE prompt stealing [51]	$2 \to 4$	I
cross-layer sources (5)		
ClawGuard [15]	$1,5 \to 1$	IV
MIRAGE [52]	$2,5 \to 2$	IV
Cloud traffic benchmark [53]	$2,5,8 \to 3$	I,II
Web-agent unmasking [54]	$3,7 \to 3,7$	II,III
LumiMAS [55]	$4,5 \to 4,6$	IV
orchestration/workflow sources (15)		
AgentPrint [4]	$5 \to 3$	I,II
Cycle detection [56]	$5 \to 4,6$	IV
SIGLEAK [57]	$5,6 \to 4,6$	I
ProcGrep [58]	$5,6 \to 4,6$	II
AgentProv [59]	$5,6 \to 6$	II
Lidar [60]	$5,6 \to 6$	II
Dangol workflow [16]	$5,6,8 \to 3,6,8$	I
Campaign attribution [61]	$5,6,8 \to 6,8$	II
WRA [6]	$5,7 \to 3$	I
PersonaFingerprint [62]	$5,7 \to 3$	I
Whose Agent? [63]	$5,7 \to 7$	II
AgentTell [64]	$5,7 \to 7$	I
UI Traces [5]	$5,7 \to 7$	II
FP-Agent [65]	$5,7 \to 7$	II
AHB [14]	$5,7 \to 7$	III

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

construction. The remaining 102 related pairs (66 traffic-envelope, 28 cache-service, three behavioral-humanization, three generator-fingerprint, one local-memory, one fixed-instruction) include the 13 same-family pairs. They show why paper-level comparison fails:

- **Same paper, different object.** Six of the 13 original same-family pairs measure different primitives, targets or cost objects, e.g., Early Bird's K-prefix oracle simulation versus its GPTCache semantic prototype, or SpliceDefense's CTBF signal suppression versus QCP-only prefill cost.
- **Same object, different population.** Of the seven aligned-outcome same-family pairs, four evaluate different collection or execution populations (e.g., Dangol's stored-capture ablation at 0.17 accuracy versus live loopback OHTTP at 0.30) and three lack a verified matched-test binding; none licenses subtraction as a paired deployment effect.
- **Across papers.** Of the 102 related pairs, 16 (15.7%) support a source-qualified descriptive comparison, 80 (78.4%) have an established receiver, target or cost-object mismatch, and 6 (5.9%) remain unresolved.

5.3. Nine-item reuse audit across 20 mitigation families

Table 8 defines each item before its application to the 180 cells in Table 9. Applicability follows the result's reuse claim: a fixed analytical attack makes adaptation inapplicable, while a component benchmark can support component utility. U and ? retain distinct source states: the 11 unverified cells are five R4 and six R5 bindings, which are distinguished from an attack failure.

TABLE 7. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [38], [41].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [16].	The co-located loopback setting yields 0.30 recovery at about 13× bytes and 11× latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are U.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [14].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [17], [35], [50].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [48].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and $<$ 5ms/$<$ 1% prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [29], [22], [18].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

SoK Takeaway 2 (Inference is bound; reuse is not). Target and inference are bound almost everywhere (R1 20/20, R7 17/20). The bindings needed to carry a result elsewhere are not: protected fit 4/11 applicable families, matched adaptation 2/11 (2/10 for learned adaptive attacks only), utility at the protected operating point 3/17 (2/16 counting task utility only) and cost 11/19. CachePrune alone binds inference, utility and cost at one operating point, and it does not establish adaptation. These are scope sensitivities, not quality ranks.

6. Systematic Holdout Evidence for Transfer

6.1. Six axes across all 82 families

Appendix D gives the complete family-by-axis matrix, including the 24 adjacent families. It codes task/input/user, model, harness/topology, hardware, network/load and time separately from the archived source-bound extraction. H denotes an explicitly reported held-out unit/domain; V a condition comparison or recalibration; X a scoped restriction; ? unresolved extraction. H does not certify every grouping detail or successful transfer, and ? is never counted as absence. Conceptual families receive an inapplicable dash.

The matrix contains H entries for 13 families on task/input/user, three on model, two on harness/topology and one on network; none is resolved as H for hardware or time. The latter axes contain seven V entries each: calibration, condition sweeps and temporal observations already exist in the literature. With 73 hardware and 71 temporal entries unresolved, this is a map of *documented evidence*, not a claim that those papers omitted such tests. Multiple H entries are counted once per family/axis, not pooled as independent experiments.

6.2. What each holdout licenses

Whisper Leak holds out target prompts; FP-Agent has a separate task holdout but unresolved visitor/host separation. Dangol's leave-composition-out and the traffic benchmark's topology shift change a different generator [41], [65], [16], [53]. LLM-Fin and ArchSnoop test unseen variants/configurations within known model families; LeakyLMs conditions architecture inference on known GPU type and calibration [72], [26], [47]. Thus a positive model holdout does not establish unknown-hardware transfer. UI Traces' recorded question overlap further shows why task labels and actual held-out units must remain distinct [5].

6.3. Operational consequence

Select the axis changed by deployment, require a source-bound separation on that axis, then check whether fitting or calibration crossed the split. Otherwise scope the result to its observed condition. Feature ablations and additional models alone supply V evidence, not a bridge to an un-

TABLE 8. OPERATIONAL RULES FOR THE NINE REUSE BINDINGS. APPLY EACH ITEM TO A SPECIFIED RESULT RECORD AND DEPLOYMENT CLAIM.

Item	Evidence required for a Bound (**B**) judgment
R1 Target	Hidden property, inference goal and authorized/protected interest are explicit.
R2 Receiver	Observer permission, released representation and auxiliary inputs identify the evaluated view.
R3 Intervention	Placement and execution mode bind the transform to stored data, wire emission or executed actions.
R4 Protected fit	Attacker fitting/calibration uses the protected condition; training and protected testing are identified.
R5 Matched adaptation	Clean- and protected-condition fits are evaluated on the same defended test population.
R6 Workload	Tasks, system/generator configuration and holdout specify the supported transfer axis.
R7 Inference	Metric, denominator and evaluation condition bind the attack outcome to the scoped mechanism.
R8 Utility	Task/component performance is measured at the same protected operating point, with its state boundary explicit.
R9 Cost	Resource units, baseline and measurement object price that operating point.

B: bound; **P**: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. First determine applicability, then assign evidence status. Family **B** means at least one scoped record satisfies the item, not that every result does.

TABLE 9. PER-FAMILY EVIDENCE BINDINGS FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [38]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	U
Early Bird [3]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	**B**
Whisper Leak [41]	**B**	**B**	**P**	?	?	**B**	**B**	U	**B**
SpliceLeak [48]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [49]	**B**	**B**	**B**	?	?	**B**	**B**	U	**B**
I Know What You Said [22]	**B**	**B**	**B**	–	–	**B**	U	–	**B**
Speculative decoding [40]	**B**	**B**	**P**	?	?	**B**	**B**	–	**B**
Dangol workflow [16]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [53]	**B**	**P**	**B**	?	?	**P**	**B**	**P**	U
Remote timing [37]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
TDXRay [29]	**B**	**B**	**B**	–	–	**B**	U	**B**	**B**
SafeKV [17]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
AHB [14]	**B**	**B**	**B**	**P**	?	**B**	**B**	**B**	U
KVGov [35]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [50]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [18]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
SparLeak [27]	**B**	**B**	**B**	?	?	**B**	**B**	**P**	U
Shape and Substance [28]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	18	17	3	11
Applicable	20	20	20	11	11	20	20	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding (11 cells); –: inapplicable. Table 8 defines R1–R9 and the shared status alphabet. A bound cell is supported by at least one scoped record. Dangol R5 concerns stored capture; its live-refit binding remains unverified. CachePrune jointly binds R7–R9 at one operating point.

seen generator. This rule also governs our endpoint experiments: within-model refits, source-trained attacker transfer and source-calibration transfer occupy separate matrix cells. The systematic result is the mismatch between the requested transfer axis and the available evidence, not a ranking by the number of holdouts.

7. Three Lenses: Where Reuse Fails

The census (§5) covers all 39 records and the holdout matrix (§6) all 82 families. We now select three examples by the *comparison decision* they expose, not by effect size: a changed receiving path, cover with different state effects, and an adaptation/cost mismatch within a named mechanism. They are explanatory, not a representative sample, and each motivates one experiment in §8.

7.1. Case 1: bind the final target-bearing emitter

The downstream re-emission boundary. NetEcho's upstream packet-size shaping operates on a different projection from the tool-call sequence. Its plaintext-aware frontend then emits a new size trace (grouped token counts or character lengths), reopening a size channel after upstream protection [38]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [34, Sections 5.1–5.3]. Both change the receiver boundary and therefore require a separately bound inference result.

The data processing inequality requires $S \to Z \to Y$: target-oblivious postprocessing cannot increase $I(S;Z)$ [73, Theorem 2.8.1]. Plaintext-aware re-emission or backward computation breaks that premise. Exp 1 measures what restoring it costs under buffered and streaming emission.

7.2. Case 2: resolve the WRA versus AHB cover asymmetry

Two axes of asymmetry: threat polarity and action risk. WRA's one-decoy E_{func} drops from 0.770 to 0.610, whereas AHB's Trip Planning success falls from 0.75 to 0.15 under `Online+Fake` humanization [6], [14]. Grouping both as "cover traffic" conceals different objectives and executions.

WRA discards decoy reads for Type I confidentiality; AHB executes GUI events for Type III evasion, potentially navigating or submitting forms. Their utility outcomes are not contradictory, and AHB's missing condition-specific denominators preclude a matched causal effect. A shared workflow setting does not establish shared state effects. Terminal-state reward and dispatch-time authorization also answer different questions: an unconfirmed write can still reach the intended final state. Exp 2 checks both.

7.3. Case 3: bind adaptive retraining and cost objects

The $D_0 \to D_2$ rebound and component-cost boundaries. A matched adaptation contrast compares clean-trained D_0 and defense-trained D_2 on identical defended tests. UI Traces and Dangol raise the record-binding question this leaves open: whether a refit, a measured delay and a utility outcome refer to the same executed condition [5], [16]. Cost objects show the same pattern. Early Bird's 6ms/4% GPTCache cost does not price K-prefix isolation; KVGov's 1.3 μs HMAC microcost omits lost reuse; SpliceDefense's QCP cost does not price CTBF+QCP [3], [35], [48]. CachePrune binds protection, QA/summarization utility and throughput, while OTRO separates conditional leakage from TTFT/padding costs [50], [18]. Exps 3–4 test calibration, refitting and timing against one cohort and matched observation sets.

8. Testing Defense Transfer Boundaries

Four experiments instantiate the boundaries of §7 using saved LLM responses and a retail agent workload: the emitter (Exp 1), executed state (Exp 2), calibration population (Exp 3) and attacker observation (Exp 4). Exps 1–2 reproduce known mechanisms to measure their streaming-cost and state/link consequences; Exps 3–4 measure agent-specific operating points. Table 10 separates populations; Appendix C indexes all studies.

Shared measurement conventions. A byte-transparent relay captures post-handshake TLS application records on loopback; TLS 1.3 session tickets are disabled. Standard-prompt results use the arm called `baseline` in the artifact; *undefended* denotes transport without a program defense. Balanced accuracy (BA) averages the two class recalls. We report the maximum over each specified attack suite and recompute that maximum in each of 2,000 common bootstrap draws, grouping tasks for reference-program tests and customers for real-agent tests. The original protocol's engineering screen requires an upper 95% BA bound below 0.60 and maximum AUROC below 0.65 for both targets. No defended configuration in Exps 3–4 passes it, so we report paired reductions against named suites rather than absolute protection (§9.3).

Attack suites. Because BA is a suite maximum, every number is indexed by its suite. The *frozen suite* (Exp 3 frozen) contains clean-trained D_0, defended-refit D_2 and training-derived decoder/count rules over lengths and order. The *11-attack union* (Exp 3 cross-fit and localization) adds learners including a 1D-CNN and a k-FP-style forest-leaf model. The *endpoint suite* (Exp 3 replications and transfer) adds a total-exchange-count rule to six attacks inherited from the frozen protocol (majority, clean- and defended-trained learners, decoded-tool and target-call-count rules). In Exp 4, the *common seven* are learners shared with the offline WF adaptations, and the *full union* (17–22 attacks) adds timing features and learned flush removal.

TABLE 10. COHORT, OBSERVATION AND COST DENOMINATORS FOR EXPS 3–4. STANDARD-PROMPT PAIRS: CANCEL/RETURN BA (%). BYTE RATIOS PRICE JOINT SHAPING AGAINST EACH STUDY'S OWN UNDEFENDED SERIALIZATION.

Study	Observation / attack suite	Undef. BA	Joint B×
Exp 3 frozen	Replay lengths/order; frozen suite	97.2/94.0	15.39
Exp 3 cross-fit	Analytic lengths/order; 11-attack union	91.7/85.5	19.76
Exp 4	New RPC envelope; lengths/order/time; full union	91.7/86.4	15.11

Frozen-split and Exp 4 rows: 276 test episodes, 24 customers, 5/7 positive customers. Exp 3 cross-fit: 690 episodes, 53 customers, 14/18 positive customers. Each prompt has half the episodes. Cross-fit fold calibration yields larger templates (5.04× calls); frozen-split and Exp 4 rows use 3.83×. Repeats measure execution variation within these customer groups.

8.1. Exp 1: bind protection and cost to the emitter

Buffered tests use brief/expanded saved responses for 24 topics (12 training, 12 test). Padding only the upstream link leaves verbosity 100% separable at a plaintext-aware terminal emitter; padding both reduces BA to 50%, reproducing NetEcho's mechanism [38]. A training-fitted buffered cap achieves 50% at 2.18× wire bytes under TLS 1.2 (2.19× under TLS 1.3).

Streaming the same 48 texts as per-token server-sent events changes the cost contract [1], [41]. Per-event padding retains token count and 100% verbosity inference. Eight-token batching cuts bytes to 0.21× and clean-trained BA to 50%, yet defended-data refitting restores 100%. Constant shape reaches 50% at 2.42× bytes and 208 versus 105 token intervals; two overflowing test outputs remain identifiable (Table 11). **Finding.** The same saved content yields different protection and delay under buffering and streaming. A buffered cap's byte ratio cannot price a streaming deployment, whose first-output delay and emission schedule are different outcomes.

8.2. Exp 2: compose state preservation with link protection

Execution boundary. Among 115 public retail tasks, 97 have an initially pending order for the task user. Three order-selection seeds yield 291 cover placements before the reference program. Same-item modification changes status and appends a refund in 173 placements despite unchanged item and price; the other 118 calls fail the variant-availability check. Idempotence, $f(f(s,a),a) = f(s,a)$, therefore differs from state preservation, $f(s,a) = s$. Cancelling another user's order changes the shared database. The shadow executor isolates cover effects on a separate state and suppresses its responses; a live-execution control exposes the resulting side effects.

The dispatch observer sees ordered tool names and response lengths. Policies add cancellation, pad its count,

TABLE 11. EXP 1: THE SAME 48 SAVED RESPONSES EMITTED BY A BUFFERED OR A TOKEN-STREAMING FRONTEND (TLS 1.3 LOOPBACK CAPTURES). BYTES ARE RELATIVE TO THE UNPADDED EMITTER OF THE SAME KIND. TTFB/DONE: FIRST AND LAST DELIVERED BYTE IN TOKEN INTERVALS (ACCOUNTING PROXY). ACCURACIES ON 24 HELD-OUT OUTPUTS (12 TOPICS); ID: CLOSED-WORLD TOP-1 IDENTIFICATION AMONG 48 RESPONSES (CHANCE 2.1%).

				Verbosity		Prefix	
Emitter / framing	Bytes×	TTFB	Done	D_0	D_2	D_2	ID
Buffered, unpadded	1.00	105	105	100.0	100.0	91.7	87.5
Buffered, training cap	2.19	105	105	50.0	50.0	50.0	2.1
Stream, per token	1.00	1	105	100.0	100.0	87.5	100.0
+ random event pad	1.29	1	105	100.0	100.0	87.9	62.3
+ constant event size	1.23	1	105	100.0	100.0	87.5	58.3
8-token batches, const.	0.21	8	105	50.0	100.0	87.5	20.8
Const. events + count cap	2.42	1	208	50.0	50.0	50.0	6.3

D_2 is the best of total-byte threshold, record-count threshold and a logistic sequence model, each fitted on defended training outputs. Random padding: 10 socket-measured draws plus an independent profiling run. TLS 1.2 gives identical accuracies and byte ratios within 0.03. Verbosity contrasts brief/expanded outputs; Prefix contrasts the first two/three sentences of each expanded output. Tokens use `o200k_base`. Training count caps are 208/88 events, with 2/24 and 5/24 test overflows, respectively.

TABLE 12. EXP 2 DISPATCH RESULTS: ALL 115 REFERENCE PROGRAMS, THREE SEEDS, KNOWN-POPULATION DEFENSE CALIBRATION; TASK-FOLDED ATTACKERS. MAXIMUM BA (CANCEL/RETURN); EXTRA CALLS PER REAL CALL; DATABASE EQUALITY WITH THE UNPROTECTED RUN.

Policy	BA (%)	Calls+	DB eq.
No cover	100.0/100.0	0.00	100%
Cancel, $p = .5$	86.1/100.0	.10	100%
Cancel, $p = 1$, shadow	100.0/100.0	.20	100%
Cancel, $p = 1$, live	100.0/100.0	.20	0%
Cancel count $\to 2$	76.9/100.0	.35	100%
One decoy program	91.9/85.9	1.01	100%
Two decoy programs	83.5/78.1	1.98	100%
Shape equalization	50.0/54.5	6.71	100%

interleave donor programs or equalize all tool counts; donors/templates use the known 115-program population. Task-grouped five-fold attacks fit clean/defended gradient boosting and logistic regression on counts, bigrams and lengths, with a count-rule baseline and three seeds (18/32 positive cancellation/return tasks). One inserted cancellation drives clean-trained BA to chance, while refitting restores 100%. Count equalization removes that rule's signal, but correlated tools retain 76.9% BA. Donor programs, inspired by WRA's discarded retrieval cover [6], leave 83.5–91.9%. Shape equalization reaches 50.0/54.5% at 6.7 extra calls per real call (Table 12). Shadow checks match the original live state in every case; live cancellation changes it in every case.

Final network view. Four cover policies and three framing choices produce 4,140 TLS 1.2 sessions. The network observer receives direction-tagged record lengths and order. A training-fold profiler maps request sizes to tool types before applying dispatch attacks: request-only identification reaches 89.9/91.4% (582 calls; majority 29.4%). Per-call buckets leave return BA at 73.9%; joint shape cover and constant framing reduce cancellation/return to 54.4/50.0%, at 26.3× bytes and 7.7× calls.

Finding. Call-level cover and per-call framing address complementary observables. Full constant-rate transport can instead hide packet counts from a passive link observer without executing dummy tools [74]; Exp 4 measures that alternative. At a dispatch view after decapsulation, cover requires hidden markers and routing to the trusted isolated executor (§9.1). The intervention must match the receiver and the protected state.

8.3. Exp 3: real-agent signatures and calibrated shaping

Task and collection. The targets are the reference program's requested operations, `cancel_pending_order` and `return_delivered_order_items`. This operational target tests whether encrypted metadata reveals which service action a task requests. GPT-6.1 Sol/low plans from public retail policies, schemas and actual sandbox observations; a separate context simulates the user. Two prompts and three repeats on 115 tasks yield 690 runs from 53 customers. The rules prompt adds a customer-details read before order access and a read before each write.

Frozen calibration and behavioral control. All 414 training runs (69 tasks) precede freezing shared cover templates, donors and framing caps. The 276 test runs cover 46 tasks from 24 disjoint customers. Prefix-only scheduling inserts training-derived cover, flushing unmet counts at episode end. Four program conditions per prompt give 2,760 local TLS replays with lengths/order attacks; every failure and handoff remains included. Real-call pass-through plus isolated cover preserves the logical response/state projection (Appendix A); reusing trajectories pairs the network interventions on identical model decisions.

Rules raise read-order adherence from 86.2% to 97.1%, while undefended category inference remains above 93%. Native rewards are 110/138 and 105/138 (paired difference -3.6 points, 95% interval $[-9.5, 1.4]$). Thus compliance and metadata privacy separate in the same task population. Joint cover and framing reach 50.0/50.0% point BA, compared with 97.2/94.0% undefended, a paired decrease of 47.2/44.0 points (Table 13). Their upper bounds (60.6/65.5%) fail the screen, and the rules-arm joint return point is 60.6%, identifying prompt/cohort sensitivity.

Customer-grouped exploratory comparison. Five-fold cross-fitting refits templates, donors, caps and all attackers on other customers, pooling the full known cohort (14/18 positive customers). The analytic TLS model matches all 2,760 saved sessions and 120 additional captures. Under the 11-attack union, joint constant shaping reduces BA by 31.2/30.0 points (paired 95% intervals [15.3,41.0]/[14.8,39.2]); buckets reduce it by 34.6/32.8 points, at lower byte cost (Table 14).

Locate the observable operation signature. An oracle selects write exchanges using true tool types, then exposes only ciphertext lengths/order. Writes comprise 19.6% of

TABLE 13. EXP 3, FROZEN CALIBRATION: OPERATION-CATEGORY INFERENCE ON REAL-AGENT TRAJECTORIES. TESTING COVERS 46 TASKS FROM 24 HELD-OUT CUSTOMERS, TWO PROMPTS AND THREE REPEATS (276 EPISODES). NETWORK BA IS THE MAXIMUM OVER THE FROZEN ATTACK SUITE, WITH CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA [95% CI]	Return BA [95% CI]	Original reward (%)	Bytes×	Calls×
Undefended	97.2 [90.0, 100.0]	94.0 [85.8, 100.0]	79.7	1.00	1.00
Rules prompt	97.2 [90.0, 100.0]	93.5 [85.0, 99.5]	76.1	1.00	1.00
Padding only	66.1 [50.0, 83.9]	55.5 [50.0, 73.0]	–	4.02	1.00
Shape only	60.8 [55.4, 71.3]	51.9 [50.5, 57.8]	–	4.16	3.83
Joint program	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	–	15.39	3.83
Rules + joint	50.0 [50.0, 65.1]	60.6 [50.0, 73.3]	–	15.72	3.88

Paired joint-minus-undefended BA differences (percentage points): standard-prompt cancellation -47.2 [-50.0, -33.3], return -44.0 [-48.9, -27.8]. These post hoc contrasts use the same customer draws and reselect the attack-suite maximum in each condition. Joint = online shadow cover plus constant framing; padding/shape ablations use standard-prompt trajectories. Ratios use undefended transport within each prompt. Shadow isolation preserves the logical planner transcript; the four program conditions replay actual LLM trajectories over local TLS 1.2. Reward is reported only for original model executions; dashes denote no new live defended-planner reward measurement (§9.3).

TABLE 14. EXP 3: EXPLORATORY CROSS-FIT COMPARISON UNDER THE 11-ATTACK UNION: ALL 690 EPISODES AND 53 CUSTOMERS. MAXIMUM NETWORK BA WITH CUSTOMER-BOOTSTRAP 95% INTERVALS. STANDARD PROMPT UNLESS MARKED RULES; BYTE/CALL RATIOS RELATIVE TO UNDEFENDED TRANSPORT.

Configuration	Cancel BA (%)	Return BA (%)	B×/C×
Undefended	91.7 [82.1, 100.0]	85.5 [76.7, 93.4]	1.0/1.0
Joint constant	60.5 [52.9, 72.4]	55.5 [51.8, 64.2]	19.8/5.0
Joint buckets	57.1 [50.0, 71.5]	52.7 [50.0, 61.7]	12.4/5.0
Rules + buckets	56.9 [50.2, 71.2]	52.7 [50.0, 65.0]	12.4/5.0

Joint constant is the frozen study's shadow-count plus constant-framing design, refit within each fold. Joint buckets changes framing to 1024-byte buckets. Every bootstrap draw reselects the complete maximum. Intervals condition on fitted predictions.

TABLE 15. EXP 3 ORACLE LOCALIZATION UNDER THE 11-ATTACK UNION. STANDARD PROMPT: 345 CROSS-FITTED EPISODES, 53 CUSTOMERS. MAXIMUM NETWORK BA (%) WITH CUSTOMER-BOOTSTRAP INTERVALS.

Target	Full trace	Writes removed	Writes only
Cancel	91.7 [82.1, 100.0]	62.8 [54.6, 75.3]	91.7 [82.1, 100.0]
Return	85.5 [76.7, 93.4]	59.9 [52.2, 71.8]	84.9 [78.8, 94.0]

TABLE 16. EXP 3 MODEL-SPECIFIC REPEATED EVALUATIONS: NETWORK SUITE-MAX BA (CANCEL/RETURN, %). EACH MODEL HAS 46 TEST TASKS AND 24 CUSTOMERS. ENDPOINT ROWS USE THE ENDPOINT SUITE; LUNA RETAINS THE FROZEN SUITE. LUNA USES 29 TRAINING TASKS; THE TWO ENDPOINT REPLICATIONS USE 69. CONTRASTS ARE WITHIN MODEL; ENDPOINT ALIASES AND INFERENCE BUDGETS DIFFER.

Model	Undefended	Joint
Luna	64.2 / 84.9	60.4 / 57.0
MiniMax-M3	58.3 / 83.5	60.4 / 67.5
GLM-5.3-flash	100.0 / 86.6	50.0 / 67.3

TABLE 17. DEFENSE CALIBRATION TRANSFER ON 46 TARGET-MODEL TEST TASKS. ENTRIES ARE CANCEL/RETURN BA (%) UNDER THE ENDPOINT SUITE; TARGET-MODEL ATTACKERS ARE REFITTED ON TRAINING CUSTOMERS. SOURCE TEMPLATES AND CAPS USE SOURCE-MODEL TRAINING DATA ONLY.

Calibration source	MiniMax target	GLM target
MiniMax	60.4 / 67.5	70.8 / 50.0
GLM	50.8 / 63.0	50.0 / 67.3

real calls yet retain 91.7/84.9% BA; removing them leaves 62.8/59.9% (Table 15). The target names an operation, so the result locates its metadata signature. It directs protection toward specific exchanges, while residual read inference shows why hiding one named call is insufficient.

Model-specific repeated evaluation. Luna's 150-run extension contributes a same-provider replication; two additional endpoints each contribute 115 standard-prompt runs with 69 training tasks, a defense frozen before test collection, and 46 test tasks (Table 16). Joint shaping lowers return BA by 16.0 points for MiniMax-M3 and 19.4 for GLM; cancellation changes by $+2.1$ and -50.0 points, respectively. Return-operation inference thus recurs under separate within-model fits, while protection depends on model and target. MiniMax-M3 is a served alias (its operator reports a DeepSeek backend); GLM uses a 16,384-token budget versus 4,096 for MiniMax; neither endpoint exposes an immutable revision.

Measure the model-transfer boundary. A post hoc transfer study keeps each target model's attackers refit on its own training customers and swaps only the defense calibration. On GLM, MiniMax calibration raises cancellation BA from 50.0% to 70.8% while lowering return from 67.3% to 50.0% (Table 17). Transfer therefore trades protection between targets; no scalar model ranking captures this boundary. Source-trained attackers also transfer: undefended return BA is 77.5% for MiniMax-to-GLM and 88.1% for GLM-to-MiniMax. These two-endpoint results use known customer partitions, not independent providers or fixed weights. Write-only padding costs 1.45/1.51× bytes for MiniMax/GLM but leaves return BA at 61.3/61.6% under the endpoint suite. A learned length-based write detector has precision 34.9/98.3% and recall 96.3/89.4%,

distinguishing oracle localization from implementable inference. Localization thus proposes economical interventions; residual signals and detector errors determine whether their protection transfers.

Test live planner utility at its own boundary. A paired 50-task GLM study runs 100 fresh planner episodes with real isolated prefix-cover execution and emulated response deadlines. Baseline and prefix-shadow succeed on 37/50 and 33/50 tasks (difference -8.0 points, customer interval [-23.9,6.5]). Withheld executed-tool responses trigger subsequent planner actions. This measures the prefix scheduler's utility under injected delays; terminal flush and TLS framing belong to a different mechanism. The supplement retains every request, fault and outcome.

Finding. Better read-order compliance leaves operation inference intact; calibrated joint shaping reduces fitted-suite inference; localization identifies the exchanges that carry it. The calibration is itself part of the defense: moved to another model, it redistributes protection between targets, so a calibrated result transfers only with its calibration population. Semantic-attribute generalization is a separate estimand (Appendix C).

8.4. Exp 4: timing, latency and matched transport controls

All 690 saved action trajectories execute through a live local tool server under six conditions, producing 4,140 new TLS sessions. Each real response gates the next action; real and shadow calls have separate states. The original 414/276 customer-disjoint split is retained, with attacks fitted and frozen before test capture. Lengths, directions and arrival times measure the tool RPC chain.

A complete tunnel sends 4,096-byte cells in both directions at 2 or 5 ms target intervals; dummy cells execute no tools. A training-sized 133-slot horizon continues after the last response. Serial and two-in-flight shadow schedules use the frozen prefix-only count policy with constant per-call framing. The full union covers size/order, timing, combined features and learned flush removal; the common seven, including a 1D-CNN and forest-leaf neighbors, permit a matched comparison with offline WF adaptations.

Timing reveals distinguishable cover. For standard-prompt serial shadow, adding timing/flush attacks to the size/order attacks increases cancellation BA from 50.0% to 61.4% (paired increase 11.4 points, 95% interval [3.7,19.6]); return changes from 50.7% to 53.4%. Flush-phase detection reaches 90.4/92.3% for serial/parallel schedules over 15,278 records from 276 test episodes. A separate cover-versus-real request classifier reaches 82.5/96.1% on 7,639 requests.

We then compose the detector with intent inference: a training-fitted detector strips the predicted suffix before classification. Table 19 shows the recovered category signal, with a strongest return point of 60.7%. The full attack union retains both stripped and unstripped traces. This separates the ability to recognize padding phases from the ability to recover a task's operation.

Match cost to the protected boundary. Within measured RPC conditions (M), per-call constant framing uses 3.92× bytes and 16.6 ms p95, versus serial shadow's 15.11× and 26.2 ms; its full-union BA is higher (68.6/62.7% versus 61.4/53.4%), a measured cost–inference trade-off. Two-in-flight shadow lowers p95 to 20.8 ms, versus 56.4/112.3 ms for 2/5 ms tunnels. Sessions preserve real responses and final live state; fixed tunnel sizes/counts still expose host-scheduled direction/order. Table 18 keeps measured RPC times separate from offline schedule transformations.

Under the common seven attacks, offline random-donor Walkie-Talkie (WT, S) reaches 50.0/56.7% BA at 2.04× bytes; measured serial shadow (M) reaches 50.0/52.9% at 15.11×. This descriptive, shared-metadata comparison identifies a lower-byte shaping candidate. Establishing its online trade-off requires RPC measurements and the extended timing/flush suite; the offline point is neither a theoretical byte lower bound nor demonstrated deployment superiority.

The WF adapters use the same saved traffic and training customers. WT samples donor traces without symmetric site pairing; WTF-PAD samples training inter-arrival histograms; Tamaraw and FRONT supply fixed-rate and front-loaded controls. Input-derived fragments preserve sizes, directions and order, enabling a controlled metadata comparison. Their simulated spans and measured RPC latency remain separate cost objects.

Emulated WAN delay and jitter. The relay re-runs all 276 test episodes with 20 ms/5 ms and 50 ms/10 ms one-way delay/jitter. The frozen loopback flush detector transfers with 89.4% BA (versus 90.4%), so the cover phase stays recognizable. Jitter does mask micro-timing: serial-shadow full-union cancellation BA falls from 61.4% to 50.0% and 54.0%. This masking depends on observer position; an observer adjacent to the tool server sees loopback-like timing. Fixed horizons fail under delay: the 2 ms tunnel's 133-slot (266 ms) horizon completes only 85/276 and 5/276 sessions, whereas shadow cover and per-call framing complete all sessions at 855.4/1,999.5 ms serial-shadow p95. A tunnel horizon must therefore be sized to path RTT, at a corresponding byte cost.

Finding. A length-only verdict misses schedule leakage. The common outcome is suite-maximum BA, always indexed by its observed features and fitted attacker set: Exp 3's 50.0/50.0% and Exp 4's 61.4/53.4% cross an attacker-observation boundary and are not interchangeable protection estimates. Within M, per-call framing costs fewer bytes and less RPC time than serial shadow but leaves higher full-union inference; S identifies lower-byte candidates under the common seven attacks. For a passive ingress-link observer, lightweight shaping is a relevant candidate; for a restricted dispatch observer, execution isolation addresses an additional state boundary. This is the audit's placement-and-cost distinction demonstrated on one workload.

Synthesis. The demonstrations operationalize the audit: match the emitter before borrowing a byte cost (Exp 1), verify executed state before borrowing a cover strategy (Exp 2), bind targets and calibration to a shared cohort (Exp 3), and price timing protection against matched trans-

TABLE 18. TIMING-AWARE EXP 4 COMPARISON: STANDARD PROMPT, 138 HELD-OUT EPISODES, 24 CUSTOMERS (5 CANCEL-POSITIVE, 7 RETURN-POSITIVE). EACH BA PAIR IS CANCEL/RETURN (%). THE COMMON SEVEN MODELS PERMIT A MATCHED-SUITE COMPARISON; THE FULL UNION ALSO RETAINS ALL EARLIER ATTACKS.

View	Condition	Common-seven BA	Full-union BA	Upper BA	Bytes×	Calls×	RPC p95 (ms)
M	Undefended	84.3/76.9	91.7/86.4	100.0/97.2	1.00	1.00	15.4
M	Per-call constant	53.2/59.1	68.6/62.7	92.8/79.5	3.92	1.00	16.6
M	Serial shadow	50.0/52.9	61.4/53.4	71.4/65.8	15.11	3.83	26.2
M	Two-in-flight shadow	50.0/57.3	51.7/57.7	69.1/69.7	15.11	3.83	20.8
M	Full tunnel, 2 ms	59.9/55.2	62.5/55.2	70.2/62.0	125.71	1.00	56.4
M	Full tunnel, 5 ms	54.6/53.7	54.6/57.3	72.9/64.2	125.71	1.00	112.3
S	Record buckets	53.1/69.8	53.1/69.8	78.5/86.9	2.25	1.00	–
S	Tamaraw 40/12 ms	55.3/55.3	55.3/55.3	68.7/64.4	12.15	1.00	–
S	Tamaraw 4/1.2 ms	57.8/51.8	57.8/51.8	71.8/61.8	11.99	1.00	–
S	FRONT	59.0/69.3	59.0/69.3	69.9/81.8	2.51	1.00	–
S	WTF-PAD adaptation	51.9/62.7	51.9/62.7	64.1/79.6	1.88	1.00	–
S	WT random-donor	50.0/56.7	50.0/56.7	67.0/67.8	2.04	1.00	–
S	FRONT, short calibration	52.4/56.0	52.4/56.0	61.3/71.1	2.33	1.00	–

M: measured local tool RPC on saved actions; S: offline traffic transformation. RPC p95 ends at the last real response, excluding LLM generation. S-row simulated spans are not RPC latency. Upper BA is the full-union 95% bound; every condition exceeds 60%. M unions contain 17–22 attacks; S unions contain the common seven. Ratios use prompt-specific undefended captures.

TABLE 19. EXP 4 INTENT BA AFTER LEARNED FLUSH REMOVAL (95% CUSTOMER-CLUSTER INTERVAL). EACH ROW: 138 TEST EPISODES, 24 CUSTOMERS. MAXIMUM OF THREE CLASSIFIERS PLUS MAJORITY; THE FULL ATTACK UNION ADDITIONALLY KEEPS UNSTRIPPED OBSERVATIONS.

Prompt	Schedule	Cancel	Return
Standard	Serial	57.8 [50.0, 70.7]	53.4 [50.0, 64.6]
Standard	Parallel	50.0 [50.0, 64.0]	50.0 [50.0, 56.7]
Rules	Serial	50.1 [50.0, 55.6]	60.7 [50.8, 70.9]
Rules	Parallel	50.8 [50.0, 61.7]	54.0 [50.0, 67.0]

port alternatives (Exp 4). They show how source-bound comparisons change a concrete deployment decision.

9. Implications, Ethics and Limits

9.1. Deployment paths and reporting obligations

A passive agent–tool observer can be an access-network operator, an authorized VPC flow monitor, or a relay on an inter-organization path. Each position defines a specific observable projection. Table 20 states the corresponding visibility and trusted endpoint. Encrypted ClientHello hides the inner server name when supported, but destination IPs and downstream connections can still identify tools [75]. Our measurements use a single ingress endpoint.

Shadow state has three possible implementations. **(1) Authorized reads and verified no-ops** use the backend with discarded responses; idempotence alone does not preserve state. **(2) Provider-supported isolation** includes Kubernetes server dry-run, separate Stripe sandboxes and transactional rollback [76], [77], [78]. These have different semantics: for example, sequence changes can survive rollback [79]. **(3) A trusted tool gateway** authenticates cover markers and routes shadow writes to an isolated executor. Our local measurements implement the gateway path; the provider-specific paths identify deployments for follow-up measurement.

TABLE 20. DEPLOYMENT BOUNDARY: THE RELEVANT PASSIVE OBSERVER AND WHAT THE MEASURED MECHANISM CAN PROTECT.

Setting	Visible information	Applicability / trusted point
Same-host sidecar	Host trace/process/IPC metadata, if permitted	Link padding alone does not hide host instrumentation; isolate the monitor's access.
Same-VPC gateway	Ingress record sizes, times and addresses	Tunnel or shadow shaping can protect this link; tunnel horizons must cover path RTT; gateway/markers remain trusted.
Remote SaaS/MCP	IP, unprotected SNI, per-endpoint flows and times	Shared ingress needed for tool-identity cover; ECH alone does not hide destination IP.
Cross-organization	Each hop's routing; provider calls/commit log	No guarantee across all hops. A trusted isolated executor is an explicit added assumption.

The trusted gateway/executor can identify cover. Protection ends at exposed markers, routing branches or commit logs; it does not protect against that principal. On the ingress link, the inexpensive WF adaptations are relevant competitors. At a restricted dispatch view, hiding call profiles additionally requires marker/routing isolation. The choice therefore follows the required observation boundary. Cover also consumes quotas and shared capacity; admission budgets, backpressure and separate credentials must prevent overload or accidental live writes.

9.2. Ethics and artifact scope

Executions use cloned public benchmark databases and loopback sockets; model requests contain public tasks and simulated interactions. No production account is modified or

third-party service probed for a vulnerability. Stakeholders include paying tenants, shared-capacity providers and people affected by state changes. Other-account cancellation is a simulator control. Released inference code can support surveillance; examples use public/synthetic identities and controlled endpoints. Bot-detection evasion has a different authorization boundary [14], [6].

AI tools assisted candidate-field extraction, implementation, source reconciliation and prose revision; the authors verified every source-linked judgment and retain responsibility for it. The artifact archives source locators, coding decisions, observations, calibration and executable checks.

9.3. Threats to validity and scope of evidence

Statistical scope. The BA/AUROC screen is an engineering threshold, not a security definition. No defended configuration meets it for both targets; paired reductions establish improvements against specified fitted suites only. Intervals reselect the suite maximum but omit retraining uncertainty. The frozen split has 5/7 positive customers; the full cohort has 14/18. Repeats add execution coverage. Reference calibration uses the known population. Real-agent calibration precedes held-out collection, but developers knew the task set; cross-fit, timing and transfer extensions are exploratory.

Targets and deployment utility. Operation targets are coupled to intended actions; localization identifies their metadata signature. The same-operation attribute diagnostic is inconclusive (Appendix C). Original agent reward, defended logical conformance and live prefix-shadow utility are distinct outcomes. Full framing/flush utility remains unmeasured. Loopback RPC timing excludes LLM generation, and WAN effects are emulated in the relay rather than measured on wide-area paths. WF adapters support matched metadata comparisons; original-protocol equivalence and equal-utility deployment comparisons remain open.

External validity. The four mutable model aliases operate in one retail environment with separate user-simulator contexts. Luna has 29 training tasks; MiniMax and GLM each have 69. Transfer matrices and live planning reuse known customers. Fresh customers, fixed weights, verified cross-provider identities, new environments and provider-backend measurements define the next validation axes.

10. Conclusion

Defense evidence for LLM and agent side channels transfers only across changes that preserve its protection and evaluation contract. At the record level, the literature binds inference far more often than the adaptation, utility and cost bindings that reuse requires; the holdout matrix makes unresolved hardware and temporal transfer explicit. Our experiments show that each unbound boundary—emitter, executed state, calibration population and attacker observation—can change a deployment decision, and that calibration transfer can exchange protection between targets rather than degrade it uniformly. The resulting procedure is simple: identify the boundary a deployment changes, require evidence bound on that boundary, and otherwise re-measure it before carrying the result forward.

References

[1] R. Weiss, D. Ayzenshteyn, G. Amit, and Y. Mirsky, "What was your prompt? a remote keylogging attack on AI assistants," in *33rd USENIX Security Symposium*. USENIX Association, 2024, pp. 3367–3384. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/weiss

[2] G. Wu, Z. Zhang, Y. Zhang, W. Wang, J. Niu, Y. Wu, and Y. Zhang, "I know what you asked: Prompt leakage via KV-cache sharing in multi-tenant LLM serving," in *Network and Distributed System Security Symposium (NDSS)*, 2025. [Online]. Available: https://www.ndss-symposium.org/ndss-paper/i-know-what-you-asked-prompt-leakage-via-kv-cache-sharing-in-multi-tenant-llm-serving/

[3] L. Song, Z. Pang, W. Wang, Z. Wang, X. Wang, H. Chen, W. Song, Y. Jin, D. Meng, and R. Hou, "The early bird catches the leak: Unveiling timing side channels in LLM serving systems," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 11 431–11 446, 2025, consulted arXiv:2409.20002v5; archived access 18 October 2026. [Online]. Available: https://ieeexplore.ieee.org/document/11206380/

[4] Y. Zhang, X. Deng, Z. Gu, Y. Chen, K. Xu, Q. Li, and J. Wu, "Exposing LLM user privacy via traffic fingerprint analysis: A study of privacy risks in LLM agent interactions," *arXiv preprint arXiv:2510.07176*, 2025. [Online]. Available: https://arxiv.org/abs/2510.07176

[5] W. Lugoloobi, S. Marro, J. Magomere, J. Wright, and C. Russell, "Known by their actions: Fingerprinting LLM browser agents via UI traces," *arXiv preprint arXiv:2605.14786*, 2026, consulted arXiv:2605.14786v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2605.14786v1

[6] H. Jeong, M. Teymoorianfard, A. Kumar, A. Houmansadr, and E. Bagdasarian, "Network-level prompt and trait leakage in local research agents," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 6485–6504. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/jeong

[7] M. Juarez, S. Afroz, G. Acar, C. Diaz, and R. Greenstadt, "A critical evaluation of website fingerprinting attacks," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 263–274.

[8] N. Mathews, J. K. Holland, S. E. Oh, M. S. Rahman, N. Hopper, and M. Wright, "SoK: A critical evaluation of efficient website fingerprinting defenses," in *IEEE Symposium on Security and Privacy*, 2023.

[9] T. A. Ghaleb, "Fingerprinting AI coding agents on GitHub," *arXiv preprint arXiv:2601.17406v1*, 2026, consulted arXiv:2601.17406v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2601.17406v1

[10] W. Wang, Z. Liu, H. Kou, X. Liu, and J. Shen, "The chatbot knows it's you: Dialogue attribution in unauthenticated human–LLM sessions," in *Proceedings of the ACM Web Conference 2026*, 2026, pp. 8797–8807. [Online]. Available: https://dl.acm.org/doi/10.1145/3774904.3793048

[11] S. Shao, Y. Li, H. Yao, Y. Chen, Y. Yang, and Z. Qin, "Reading between the lines: Towards reliable black-box LLM fingerprinting via zeroth-order gradient estimation," 2026, version 2, 21 January 2026. [Online]. Available: https://arxiv.org/html/2510.06605v2

[12] J. Leshin, M. Shah, I. Timmis, and D. Kang, "Behavioral fingerprints for LLM endpoint stability and identity," 2026, version 1, 19 March 2026. [Online]. Available: https://arxiv.org/html/2603.19022v1

[13] T. Zhu, Y. Wu, W. Jin, Y. Li, and G. Huang, "StepFinder: A temporal semantic framework for failure attribution in multi-agent systems," 2026, version 1, 2 June 2026. [Online]. Available: https://arxiv.org/html/2606.03467v1

[14] J. Zhu, L. Yang, R. Shan, C. Zheng, Z. Zheng, W. Liu, Y. Yu, W. Zhang, and J. Lin, "Turing test on screen: A benchmark for mobile GUI agent humanization," *arXiv preprint arXiv:2604.09574v1*, 2026, consulted arXiv:2604.09574v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/html/2604.09574v1

[15] L. L. Gan, J. Wu, L. Ge, L. Yang, Y. Song, J. Zhang, H. Jin, W. Wang, and G. Xue, "ClawGuard: Out-of-band detection of LLM agent workflow hijacking via EM side channel," *arXiv preprint arXiv:2605.06205v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.06205v1

[16] B. Dangol, "From privacy to workflow integrity: Communication-graph metadata in autonomous agent interoperability," *arXiv preprint arXiv:2606.07150v3*, 2026, consulted arXiv:2606.07150v3; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2606.07150v3

[17] K. Chu, Z. Lin, D. Xiang, Z. Shen, J. Su, C. Chu, Y. Yang, W. Zhang, W. Wu, and W. Zhang, "Selective KV-cache sharing to mitigate timing side-channels in LLM inference," *arXiv preprint arXiv:2508.08438v2*, 2026, consulted arXiv:2508.08438v2; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2508.08438v2

[18] J. Lee, Y. Wang, R. Rajat, D. Wong, M. Li, and M. Annavaram, "Otro: Oblivious tokenization path with square-root oram," *arXiv preprint arXiv:2606.17358v1*, 2026, consulted 2606.17358v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.17358v1

[19] N. Alhalawani, G. Acar, Z. Liu, and L. Batina, "Physical side-channel attacks on edge LLMs: Extracting output tokens and user prompts," *arXiv preprint arXiv:2610.14215v1*, 2026, consulted 2610.14215v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.14215v1

[20] A. Chaudhuri, S. Shukla, S. Bhattacharya, and D. Mukhopadhyay, "Energon: Unveiling transformers from GPU power and thermal side-channels," in *IEEE/ACM International Conference on Computer-Aided Design*, 2025. [Online]. Available: https://arxiv.org/abs/2508.01768v1

[21] R. Ding, T. Xu, X. Shen, A. A. Ding, and Y. Fei, "MoEcho: Exploiting side-channel attacks to compromise user privacy in mixture-of-experts LLMs," in *ACM SIGSAC Conference on Computer and Communications Security*, 2025, pp. 2159–2173. [Online]. Available: https://arxiv.org/abs/2508.15036v1

[22] Z. Gao, J. Hu, F. Guo, Y. Zhang, Y. Han, S. Liu, H. Li, and Z. Lv, "I know what you said: Unveiling hardware cache side-channels in local large language model inference," in *34th USENIX Security Symposium*, 2025, pp. 1649–1668. [Online]. Available: https://www.usenix.org/conference/usenixsecurity25/presentation/gao-zibo

[23] C. Gu, R. Levine, Z. Zhang, T. Sorensen, and Y. Guo, "Behind bars: A side-channel attack on NVIDIA MIG cache partitioning using memory barriers," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 5373–5392. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/gu-cheng

[24] R. Nazaraliyev, Y. Zhang, and N. Abu-Ghazaleh, "Page404: Breaking GPU isolation through hardware-assisted page migration," *arXiv preprint arXiv:2610.20962v1*, 2026, consulted 2610.20962v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.20962v1

[25] A. Adiletta and B. Sunar, "Spill the beans: Exploiting CPU cache side-channels to leak tokens from large language models," *arXiv preprint arXiv:2505.00817*, 2025. [Online]. Available: https://arxiv.org/abs/2505.00817

[26] H. Weng, R. Zhou, Y. Qu, X. Ji, and W. Xu, "ArchSnoop: LLM architecture snooping via electromagnetic side-channel on edge devices," in *Proceedings of the 19th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, 2026, pp. 51–62. [Online]. Available: https://dl.acm.org/doi/10.1145/3765613.3811673

[27] F. Chen, L. Du, J. Zhou, P. Li, and Z. Su, "Sparleak: Privacy leakage from sparse attention in LLM inference on shared gpus," *arXiv preprint arXiv:2609.38830v1*, 2026, consulted 2609.38830v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.38830v1

[28] E. Hadad and M. Guri, "Shape and substance: Dual-layer side-channel attacks on local vision-language models," *arXiv preprint arXiv:2603.25403v1*, 2026, consulted 2603.25403v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2603.25403v1

[29] T. Hornetz, H. Yavarzadeh, A. Cheu, A. Gascon, L. Gerlach, D. Moghimi, P. Schoppmann, M. Schwarz, and R. Zhang, "Tdxray: Microarchitectural side-channel analysis of Intel TDX for real-world workloads," in *IEEE Symposium on Security and Privacy*, 2026. [Online]. Available: https://tdxray.cpusec.org/assets/tdxray_sp26.pdf

[30] Y. Jo, J. Park, E. Lee, and D. Song, "Sparseety: Extracting tokens from sparsity-exploiting LLM serving systems via deterministic side channels," *arXiv preprint arXiv:2608.02995v1*, 2026, consulted 2608.02995v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2608.02995v1

[31] R. Weiss, B. Konstantinov, E. Sheetrit, T. Simon, and Y. Mirsky, "Detokenization leaks: Reconstructing local LLM outputs from cache traces," *arXiv preprint arXiv:2609.06674v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.06674v1

[32] J.-J. Jiao, H. Yang, and R. Wen, "Llm-wfin: A fine-grained large language model (LLM)-oriented website fingerprinting attack via fusing interrupt trace and network traffic," *Electronics*, vol. 14, no. 7, p. 1263, 2025, consulted publisher version 2025-03-23; source-scoped access 21 October 2026. [Online]. Available: https://doi.org/10.3390/electronics14071263

[33] D. Scheinert, A. Acker, T. Wittkopp, S. Becker, H. Yous, K. Reddy, I. Farhat, H. Hacid, and O. Kao, "Beyond microservices: Testing web-scale RCA methods on GPU-driven LLM workloads," 2026, version 1, 2 March 2026. [Online]. Available: https://arxiv.org/html/2603.02057v1

[34] G. Politis and E. Pappas, "Privacy failure in split-LLM training, the returned gradient nullifies the decoys," *arXiv preprint arXiv:2609.04382v1*, 2026, consulted 2609.04382v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.04382v1

[35] T. C. Addagada, "Governing the KV cache: Preventing timing side-channel leakage in multi-tenant LLM inference," *arXiv preprint arXiv:2608.09225*, 2026, consulted arXiv:2608.09225v1; archived access 19 October 2026. [Online]. Available: https://arxiv.org/abs/2608.09225v1

[36] S. Alhazbi, A. Hussain, G. Oligeri, and P. Papadimitratos, "LLMs have rhythm: Fingerprinting large language models using inter-token times and network traffic analysis," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 5050–5071, 2025. [Online]. Available: https://arxiv.org/abs/2502.20589v1

[37] N. Carlini and M. Nasr, "Remote timing attacks on efficient language model inference," *arXiv preprint arXiv:2410.17175v1*, 2024. [Online]. Available: https://arxiv.org/abs/2410.17175v1

[38] Z. Zhang, G. Wu, S. Deng, S. Wang, and Y. Zhang, "Netecho: From real-world streaming side-channels to full LLM conversation recovery," *arXiv preprint arXiv:2510.25472*, 2025, consulted arXiv:2510.25472v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2510.25472v1

[39] S. Li, T. Cui, M. Chen, X. Lin, Z. Gu, X. Deng, K. Xu, and Q. Li, "From length to content: Token-length side-channel attacks on LLM api merged outputs," in *35th USENIX Security Symposium*, 2026, pp. 2187–2206. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/li-sijia

[40] J. Wei, A. Abdulrazzag, T. Zhang, A. Muursepp, and G. Saileshwar, "When speculation spills secrets: Side channels via speculative decoding in llms," *arXiv preprint arXiv:2411.01076*, 2024, consulted arXiv:2411.01076v4; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2411.01076v4

[41] G. McDonald and J. B. Or, "Whisper leak: a side-channel attack on large language models," *arXiv preprint arXiv:2511.03675*, 2025, consulted arXiv:2511.03675v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2511.03675v1

[42] R. A. Bakar, "Characterizing contention-induced reliability collapse in KV-cache timing side channels for multi-tenant LLM serving," *arXiv preprint arXiv:2609.06853v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.06853v1

[43] T.-R. Chen, "Sensitive-topic leakage through LLM routing metadata: Measurement and mitigation," *arXiv preprint arXiv:2610.09981v1*, 2026, consulted 2610.09981v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.09981v1

[44] E. Debenedetti, G. Severi, N. Carlini, C. A. Choquette-Choo, M. Jagielski, M. Nasr, E. Wallace, and F. Tramèr, "Privacy side channels in machine learning systems," in *33rd USENIX Security Symposium*. USENIX Association, 2024. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/debenedetti

[45] C. Gu, X. L. Li, R. Kuditipudi, P. Liang, and T. Hashimoto, "Auditing prompt caching in language model apis," in *International Conference on Machine Learning*, vol. 267, 2025, pp. 20 477–20 496. [Online]. Available: https://proceedings.mlr.press/v267/gu25b.html

[46] X. Zheng, H. Han, S. Shi, Q. Fang, Z. Du, X. Hu, and Q. Guo, "Inputsnatch: Stealing input in LLM services via cache-sharing timing side-channel attacks," in *Advanced Parallel Processing Technologies (APPT 2026)*, 2027, pp. 483–497, first online August 24, 2026; publisher citation year 2027. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-92-4805-6_32

[47] S. Majidi, N. Mireshghallah, and K. Taram, "Leaky language models: Stealing architecture and inference optimizations via per-token timing," *arXiv preprint arXiv:2607.20723*, 2026, consulted arXiv:2607.20723v1; archived access 19 October 2026. [Online]. Available: https://arxiv.org/abs/2607.20723v1

[48] H. Sun, S. Liu, S. Ma, J. Li, M. Xiao, and W. Jiang, "Agent-assisted side-channel attacks on non-prefix KV cache in rag," *arXiv preprint arXiv:2606.21842*, 2026, consulted arXiv:2606.21842v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2606.21842v1

[49] T. Zhang, G. Saileshwar, and D. Lie, "Time will tell: Timing side channels via output token count in large language models," *arXiv preprint arXiv:2412.15431*, 2024, consulted arXiv:2412.15431v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2412.15431v1

[50] G. Wu, Z. Li, Y. Zhang, Z. Zhang, J. Niu, Y. Wu, and Y. Zhang, "Cacheprune: Privacy-aware and fine-grained KV cache sharing for efficient LLM inference," *arXiv preprint arXiv:2605.23640v1*, 2026, consulted 2605.23640v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2605.23640v1

[51] I. Yona, I. Shumailov, J. Hayes, and N. Carlini, "Stealing user prompts from mixture of experts," *arXiv preprint arXiv:2410.22884v1*, 2024. [Online]. Available: https://arxiv.org/abs/2410.22884v1

[52] P. Revankar, K. Chauhan, J. Kim, S. N. Nur, V. Siu, and C.-G. Wang, "Mirage: A polarity-flipping encoding subspace in LLM agents," *arXiv preprint arXiv:2606.10304v1*, 2026, consulted 2606.10304v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.10304v1

[53] S. Pouryousef, J. Lopez, S. R. Nowmi, M. M. Kamol, M. Hossain, M. Tran, and M. S. Rahman, "A large-scale benchmark and risk assessment of traffic analysis attacks on cloud LLM services," *arXiv preprint arXiv:2609.31877v1*, 2026, consulted arXiv:2609.31877v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2609.31877v1

[54] I. Fayolle, S. Bouhenniche, S. Pélissier, P. Laperdrix, C. Maurice, and W. Rudametkin, "On the internet, nobody knows you're an LLM bot: Unmasking web agents with multi-layer fingerprinting," *arXiv preprint arXiv:2606.30119v1*, 2026, consulted 2606.30119v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.30119v1

[55] R. Solomon, Y. Y. Levi, L. Vaknin, E. Aizikovich *et al.*, "LumiMAS: A comprehensive framework for real-time monitoring and enhanced observability in multi-agent systems," *arXiv preprint arXiv:2508.12412v2*, 2026. [Online]. Available: https://arxiv.org/abs/2508.12412v2

[56] F. George, H. Kumar, D. Pathak, K. Ray, M. Verma, and P. Moogi, "Unsupervised cycle detection in agentic applications," *arXiv preprint arXiv:2511.10650v1*, 2025. [Online]. Available: https://arxiv.org/abs/2511.10650v1

[57] J. Geng, R. He, Z. Fei, B. Yi, R. Wang, Z. Liu, X. Hu, X. Wu, and Q. Zeng, "Agent skills matter: Inferring proprietary skills from execution trajectories," *arXiv preprint arXiv:2607.25560v1*, 2026, consulted arXiv:2607.25560v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2607.25560v1

[58] H. Oderinwale, "Agent trajectories as programs: fingerprinting and programming coding-agent behavior," *arXiv preprint arXiv:2606.16988v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.16988v1

[59] X. Wang, B. Zhao, M. Backes, F. Boenisch, and A. Dziedzic, "Agentprov: Auditing agentic LLM api providers via tool-use policy probes," *arXiv preprint arXiv:2609.00052v1*, 2026, consulted 2609.00052v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.00052v1

[60] C.-Y. Wang, X.-H. Xie, T.-Z. Wang, F.-C. Luo, and Y. Cui, "Who is behind the harness? fingerprinting LLMs through agentic behavior," *arXiv preprint arXiv:2609.28559v1*, 2026, consulted 2609.28559v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.28559v1

[61] S.-J. Park, M. Choi, J. Kim, and M.-S. Kang, "Cross-agent campaign attribution: Linking asynchronous attacks across LLM agents," *arXiv preprint arXiv:2607.18826v1*, 2026, consulted 2607.18826v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2607.18826v1

[62] C. Song, H. Wang, and R. Martin, "Personafingerprint: Measuring persona inference on modern websites with LLM-driven browsing," *arXiv preprint arXiv:2605.15962*, 2026. [Online]. Available: https://arxiv.org/abs/2605.15962

[63] D. Kang, H. Jeong, J. Sheffey, P. Datta, and A. Houmansadr, "Whose agent are you? multi-layer fingerprinting and attribution of autonomous web agents," *arXiv preprint arXiv:2606.20910v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.20910v1

[64] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "AgentTell: Behavioural side-channel leakage in browser-use agents," *arXiv preprint arXiv:2609.32915v1*, 2026, consulted arXiv:2609.32915v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2609.32915v1

[65] E. Wang, Z. Shafiq, and Y. Vekaria, "Fp-agent: Fingerprinting AI browsing agents," *arXiv preprint arXiv:2605.01247*, 2026, consulted arXiv:2605.01247v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2605.01247v1

[66] S. Banerjee, P. Sahu, M. Luo, A. Vahldiek-Oberwagner, N. J. Yadwadkar, and M. Tiwari, "SoK: A systems perspective on compound AI threats and countermeasures," *arXiv preprint arXiv:2411.13459v1*, 2024, consulted arXiv:2411.13459v1; inspected 20 October 2026. [Online]. Available: https://arxiv.org/abs/2411.13459v1

[67] J. Forough, M. Kogias, and H. Haddadi, "When agents handle secrets: A survey of confidential computing for agentic AI," *arXiv preprint arXiv:2605.03213v2*, 2026, consulted arXiv:2605.03213v2; inspected 20 October 2026. [Online]. Available: https://arxiv.org/abs/2605.03213v2

[68] O. Kotevska, S. Jha, A. Bellet, R. Hu, N. D. Bastian, R. F. da Silva, R. Madduri, and K. Kim, "Privacy foundations for multi-institutional scientific artificial intelligence," *arXiv preprint arXiv:2609.39787v1*, 2026, consulted 2609.39787v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.39787v1

[69] X. Cai, R. Nithyanand, T. Wang, R. Johnson, and I. Goldberg, "A systematic approach to developing and evaluating website fingerprinting defenses," in *ACM CCS*, 2014, pp. 227–238. [Online]. Available: https://doi.org/10.1145/2660267.2660362

[70] M. Juarez, M. Imani, M. Perry, C. Diaz, and M. Wright, "Toward an efficient website fingerprinting defense," in *ESORICS*, 2016, pp. 27–46. [Online]. Available: https://arxiv.org/abs/1512.00524

[71] T. Wang and I. Goldberg, "Walkie-talkie: An efficient defense against passive website fingerprinting attacks," in *26th USENIX Security Symposium*, 2017. [Online]. Available: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/wang-tao

[72] N. Nazari, F. Xiang, C. Fang, H. M. Makrani, A. Puri, K. Patwari, H. Sayadi, S. Rafatirad, C.-N. Chuah, and H. Homayoun, "LLM-FIN: Large language models fingerprinting attack on edge devices," in *International Symposium on Quality Electronic Design*, 2024. [Online]. Available: https://web.cs.ucdavis.edu/~srafatir/pages/documents/llm_fingerprint.pdf

[73] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006, theorem 2.8.1, Data-processing inequality. [Online]. Available: https://onlinelibrary.wiley.com/doi/book/10.1002/047174882X

[74] C. Hopps, "Aggregation and Fragmentation Mode for Encapsulating Security Payload (ESP) and Its Use for IP Traffic Flow Security (IP-TFS)," RFC Editor, Tech. Rep. RFC 9347, Jan. 2023. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9347

[75] E. Rescorla, K. Oku, N. Sullivan, and C. A. Wood, "TLS Encrypted Client Hello," RFC 9849, Mar. 2026. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9849

[76] Kubernetes Authors, "Kubernetes API Concepts: Dry-run," https://kubernetes.io/docs/reference/using-api/api-concepts/#dry-run, accessed October 24, 2026.

[77] Stripe, "Sandboxes," https://docs.stripe.com/sandboxes, accessed October 24, 2026.

[78] PostgreSQL Global Development Group, "PostgreSQL 18 Documentation: ROLLBACK," https://www.postgresql.org/docs/18/sql-rollback.html, accessed October 24, 2026.

[79] ——, "PostgreSQL 18 Documentation: Sequence Manipulation Functions," https://www.postgresql.org/docs/18/functions-sequence.html, accessed October 24, 2026.

[80] J. Roh, E. Bagdasarian, H. Haddadi, and A. S. Shamsabadi, "SPILLage: Agentic oversharing on the web," *arXiv preprint arXiv:2602.13516v1*, 2026. [Online]. Available: https://arxiv.org/abs/2602.13516v1

[81] G. Yu, Q. Wang, R. Lang, S. Su, and X. Wang, "PlanTwin: Privacy-preserving planning abstractions for cloud-assisted LLM agents," *arXiv preprint arXiv:2603.18377v2*, 2026. [Online]. Available: https://arxiv.org/abs/2603.18377v2

[82] J. Xu, L. Fan, Z. Wang, X. Li, and H. Liu, "Selection-invariant communication compilers for privacy-aware multi-agent LLM workflows," *arXiv preprint arXiv:2609.26076v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.26076v1

[83] Y. Qiao, D. Liu, H. Yang, W. Zhou, and S. Hu, "Agent tools orchestration leaks more: Dataset, benchmark, and mitigation," *arXiv preprint arXiv:2512.16310v4*, 2026, version 4. [Online]. Available: https://arxiv.org/abs/2512.16310v4

[84] M. Umar, A. P. Marathe, M. D. Gupta, S. J. Ghosh, G. E. Suh, and W. Xiong, "Efficient memory side-channel protection for embedding generation in machine learning," in *IEEE International Symposium on High-Performance Computer Architecture*, 2025. [Online]. Available: https://bearhw.ece.vt.edu/content/dam/bearhw_ece_vt_edu/publications/2025_HPCA_SecEmb.pdf

[85] L. Wu, C. Wang, T. Liu, Y. Zhao, and H. Wang, "From assistants to adversaries: Exploring the security risks of mobile LLM agents," *arXiv preprint arXiv:2505.12981v1*, 2025, version 1. [Online]. Available: https://arxiv.org/abs/2505.12981

[86] M. Mudryi, M. Chaklosh, and G. M. Wójcik, "The hidden dangers of browsing AI agents," *arXiv preprint arXiv:2505.13076v1*, 2025, version 1. [Online]. Available: https://arxiv.org/abs/2505.13076

[87] A. Ukani, H. Haddadi, A. S. Shamsabadi, and P. Snyder, "Privacy practices of browser agents," *arXiv preprint arXiv:2512.07725*, 2025. [Online]. Available: https://arxiv.org/abs/2512.07725

[88] R. H. Anwar, J. O. Weiss, T. Alves, S. Kundu, and M. T. Raza, "InferNet: Exploiting aggregate GPU profiles as side-channel for DNN architecture inference," *ACM Transactions on AI Security and Privacy*, 2026. [Online]. Available: https://rhasnainanwar.me/assets/pdf/2304.03388v2.pdf

[89] W. Cui, H. Zhang, Y. Huang, Q. Li *et al.*, "SymDiag: Explainable diagnosis for LLM reasoning via neuro-symbolic verification," in *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, 2026, consulted arXiv:2608.08786v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2608.08786v1

[90] N. Lahjouji and A. G. Colaco, "Agents that know too much: A data-centric survey of privacy in LLM agents," *arXiv preprint arXiv:2606.26627*, 2026, consulted arXiv:2606.26627v1; archived access 19 October 2026. [Online]. Available: https://arxiv.org/abs/2606.26627v1

[91] J. W. Kim, B. Min, K. Edemacu, Y. Choi, S.-H. Cho, and B. Jang, "Dp-memview: A memory interface for attribute-level transcript privacy in long-term LLM agents," *arXiv preprint arXiv:2608.03130v1*, 2026, consulted 2608.03130v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2608.03130v1

[92] Y.-M. Wang, Y.-X. Li, S.-L. Fan, Y.-H. Zheng, and T.-H. Chang, "Ask without telling: Local slms consult cloud LLMs without revealing task intent," *arXiv preprint arXiv:2609.32642v1*, 2026, consulted 2609.32642v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.32642v1

[93] Z. Liu, S. van Hoek, P. Horváth, D. Lauret, X. Xu, and L. Batina, "Real-world edge neural network implementations leak private interactions through physical side channel," *arXiv preprint arXiv:2501.14512v1*, 2025, consulted 2501.14512v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2501.14512v1

[94] P. G. Pennas, K. Papaioannou, M. Guarnieri, and T. D. Doudali, "Cachesolidarity: Preventing prefix caching side channels in multi-tenant LLM serving systems," *arXiv preprint arXiv:2603.10726v1*, 2026, consulted 2603.10726v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2603.10726v1

[95] C.-Y. Zhou, Q.-L. Jiang, S.-N. Wu, and X.-H. Zhou, "The vision encoder as a privacy boundary: Visual-token side channels in encoder-free vision-language models," *arXiv preprint arXiv:2606.14783v1*, 2026, consulted 2606.14783v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.14783v1

[96] G. Politis and E. Pappas, "What gradients add to text leakage in split language models, counted per token and per document," *arXiv preprint arXiv:2610.04128v1*, 2026, consulted 2610.04128v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.04128v1

Appendix A.
Audit Trail and Logical Equivalence

Evidence organization. The package separates the 82-family source map, 39-result ledger, 180-cell audit, transport tests, dispatch tests and real-agent studies. Each source

record has a version and section/table/figure/paragraph locator. The supplement supplies detailed algorithms, full result grids and archived coding decisions; the main paper contains the comparison rules, complete family matrix and record index needed to follow the synthesis.

Logical equivalence. For deterministic tools, a real action gives $(s_{t+1}, o_t) = T(s_t, a_t)$. The defended dispatcher forwards that action unchanged, runs cover on a disjoint state s_t^{sh} and suppresses cover responses. With equal initial states and realized real actions, induction gives

$$s_t^D = s_t, \qquad \pi_{\mathrm{plan}}(D(H_t)) = H_t.$$

Here H_t is the logical message/action/response history, excluding wall-clock time. Payload-preserving framing and terminal shadow flush maintain the invariant, provided timeouts, shared-state side effects and timing-dependent tool behavior are absent. Checks compare every real-call projection and scoring field across all 690 trajectories and four program conditions. These establish implementation conformance to isolation. The utility boundary in §9.3 explains the further live-planner measurements.

The reproducibility index resolves each archived experiment in Appendix C to protocol, inputs, predictions and checksums. Saved-output verification reproduces the reported metrics without new model calls, while measured captures and analytic reconstructions retain their separate provenance.

Appendix B. Result-record index

Table 21 indexes all 39 records with their primary measurement mode. Separate records identify changes in observation, intervention, target, execution population or cost object within a family. The six contextual records (‡) and the six records added by the final search (†) are marked; the remaining 27 are the mitigation records of the initial ledger used by the pairwise audit (§5). Shortened source locators support direct lookup; the machine-readable ledger retains exact versions, hashes and full locators.

Appendix C. Experiment artifact index

Table 22 maps the four main experiments to stable artifact keys and detailed supplementary methods. Reanalysis, prompts and repetitions retain their original customer/task membership.

The reproducibility index resolves these keys to files, protocols, saved predictions and executable verification commands. The source versions and calibration chronology remain available for every analysis.

Extensions and sensitivity. The same-provider Luna extension contains 150 formal runs on 75 existing retail tasks, two prompts and one repeat. Standard-prompt write-only dispatch BA is 87.5/66.6%, versus 50.0/57.0% for get/find/list reads; rewards are 32/75 and 33/75, with seven protocol-error episodes retained. This tests the operation diagnostic under another model alias.

A customer-composition check on fixed cross-fit predictions changes standard-prompt joint-constant BA from 60.5/55.5% to 59.0/53.2% under equal-customer-within-class weighting. Rules-arm return changes from 59.4% to 60.4%. Across 192 leave-one-positive-customer checks, standard-prompt cancellation ranges 57.3–62.0%; these are influence ranges, not confidence intervals.

The same-operation attribute check fixes single-item returns and a pre-fit digital-device grouping. Per prompt, 20 customer–order–item instances from 15 customers use the original customer folds and fixed LR/RF/GB learners. Maximum OOF BA is 30.3% [16.7,52.7], versus 50% for a constant predictor; a single fixed shuffled target gives 65.2%. Both prompts have identical features. Below-chance predictions can reflect reversed associations, and one shuffle is not a permutation distribution. This diagnostic is inconclusive for non-action attributes; its full predictions remain in the supplement.

Appendix D. Family-Level Holdout Evidence

Tables 23–25 cover all 82 indexed families. T: task/input/user; M: model; C: harness/topology; H: hardware; N: network/load; D: time. **H** records an explicitly reported held-out unit/domain, **V** a condition comparison or recalibration, **X** an explicit restriction or absent/unverified holdout in the inspected scope, and **?** an unresolved axis. A dash applies only to conceptual families. H does not certify all grouping details or successful transfer; V is not a holdout. Notes restrict each code. Core and adjacent families remain separate (C/A). The machine-readable matrix preserves the full source excerpts, locators and input hash; these tables synthesize that archived extraction rather than claim a new full-paper audit.

TABLE 21. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). MODE IS THE PRIMARY MEASUREMENT MODE: EX EXECUTED COLLECTION/PROTOTYPE (20), TR TRACE TRANSFORMATION (6), SI SIMULATION/DERIVED ESTIMATE (5), CO COMPONENT/CALIBRATION MEASUREMENT (6), UN PLACEMENT UNRESOLVED (2). ‡: CONTEXTUAL RECORD (6); †: ADDED BY THE FINAL SEARCH (6). T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

Record ID / family	Mode	Boundary; source locator
UI-delay-offline [5]	Tr	Offline delay; no live task outcome. 6.1 F5; A.5
WRA-decoy-execution [6]	Ex	Executed decoys; usefulness proxy only. 8.1–8.3 T8–11
NET-upstream-local [38]	Ex	Protected deployment/local serving; topology scoped. IV-C–E TIII–IV
NET-dummy-padding-simulation [38]	Tr	Synthetic insert/pad; no live execution. VII TIX
EARLY-K-calibration-oracle [3]	Si	K-specific fit then oracle; not live KV trial. V-A TVI; IV-A
EARLY-semantic-deidentification [3]	Ex	GPTCache prototype cost; not KV-granularity cost. V-B
WHISPER-randompadding-capture [41]	Ex	New padded captures; rounded medians. 6.1 T3–4
WHISPER-batching-replay [41]	Tr	Offline batching; model-dependent result. 6.2 F7
WHISPER-injection-replay [41]	Tr	Synthetic packets; no live zero-delay claim. 6.3 T5
SPLICE-CTBF-signal [48]	Ex	Prototype background load; tested signal only. VII-C TII
SPLICE-QCP-cost [48]	Co	QCP component cost; excludes combined CTBF. VII-C TII
TIME-length-instruction [49]	Ex	Length instruction; unequal outputs remain. 7.2
TIME-padding-estimate [49]	Si	Analytic plaintext cost; not network bytes. 7.3
SPEC-padding-REST [40]	Un	Padding/aggregation; payload cost, placement unverified. 6.1 F6–7 T2
DANGOL-capture [16]	Tr	Capture transform; distinct from live population. 9.8–9.9 T6
DANGOL-livebinding [16]	Ex	Live loopback binding; do not join capture accuracy. 9.7, OHTTP paragraph after T5
SIG-common-guardrail‡ [57]	Ex	Common instruction; not a causal guardrail effect. 4.1–4.2; App. C
SAFE-post-trigger [17]	Ex	Post-trigger probes; detector cost not service TTFT. 7.2.4 F12; 7.3–7.4
AHB-online-executed [14]	Ex	Executed UI edits; task/state changes possible. 5.1.2 T1; E.1–E.3
AHB-offline [14]	Tr	Offline touch edits; no online task damage claim. T1 offline rows; E.1–E.3
KVGOV-partialflush-sim [35]	Si	Deterministic simulation; not live protected recovery. 5.1–5.3
KVGOV-hardware-gap [35]	Co	Hardware gap calibration; not protected ASR. 5.7–5.8
KVGOV-HMAC-cost [35]	Co	HMAC microcost; not cache reuse/task loss. 5.9
TRAFFIC-prompt-reformulation [53]	Ex	Collected prompt rewrites; task utility unmeasured. V-C1 F9a
TRAFFIC-temperature [53]	Ex	Collected decoding changes; no monotone benefit. V-C1 F9a
CARLINI-fixed-rate [37]	Un	Rate/replayed cost; per-token, not request delay. VII F11–12
GAO-zero-copy-off [22]	Co	Loader component costs; not end-to-end TTFT. 7.1
TDXRAY-oblivious-map [29]	Co	Lookup microbenchmark; not full inference cost. 6.1 F8 T3
AGENTTELL-fixed-instruction‡ [64]	Ex	Live local browser tasks; loaded/cold context contrast. 4.3.1; 5 Finding 1
FP-CLOUDFLARE‡ [65]	Ex	Live access control; not user confidentiality. 6.4 T5
BAR-co-running-noise‡ [23]	Ex	MIG noise test; not an LLM defense trial. 7.2–7.3 T8
RHYTHM-VPN‡ [36]	Ex	Live VPN shift; baseline-trained attacker only. VI-B; VII
DEB-filter-origin‡ [44]	Ex	Filter creates attack view; not mitigation of it. Threat model; 4.2
CACHEPRUNE-selective† [50]	Ex	Selective cache sharing; direct token recovery/quality. 3.1; 5.1–5.6; T1; App. A
CACHEPRUNE-context† [50]	Ex	Contextual guessing; distinct from direct recovery. 3.1; 5.1–5.6; T1; App. A
OTRO-unpadded† [18]	Co	ORAM TTFT/memory; access-count padding excluded. VII-C–E; TIII
OTRO-epochpad† [18]	Si	Epoch-padding estimate; not measured padded TTFT. VII-C–E; TIII
SPARLEAK-replacement† [27]	Ex	Sparse replacement/quality; attack cost is separate. 3.1; 4–7; T4
SHAPE-padding-estimate† [28]	Si	Constant-work proposal/estimate; no executed pair. IX-B1

TABLE 22. FOUR MAIN-TEXT EXPERIMENTS AND THEIR ARTIFACT KEYS. THE RIGHT COLUMN GIVES THE SUPPLEMENT SECTION WITH IMPLEMENTATION AND FULL-GRID DETAIL.

Experiment	Artifact keys	Observation/execution boundary and comparison	Detailed source
Exp. 1: transport emission	E1, E3, E5	Buffered two-link re-emission and token-stream emitters; upstream framing, final-link framing and matched attacker refitting are distinct comparisons.	S9/S11; S12
Exp. 2: execution and link	E2, E6, E7	Prescribed stateful plans and randomized dispatch cover, then agent–tool TLS composition; dispatch-visible calls and encrypted-link records are separate views.	S9/S11; S12
Exp. 3: real trajectories and localization	E9, E10, E12	Saved autonomous Sol trajectories; original pre-test calibration, exploratory customer cross-fitting and oracle operation-category localization retain their different fit/test boundaries.	S14; S15/S21
Exp. 4: time and WF	E13, E14	Local live tool-loop timing/full tunnels, then stronger learners and specified WF adaptations; measured clocks and simulations are distinguished, with matched suites for joint comparisons.	S19; S20
Extensions	E8	Stand-in-planner pipeline check.	S13
	E11	Retrospective training-fold operating-point selection with the historical suite.	S15
	E15	Additional same-provider Luna cohort.	S22
	E16	Same-operation semantic-attribute diagnostic on reused Sol data.	S23.6
	R11/R12	Two served model endpoints: native trajectories, frozen training calibration and TLS write localization.	S24
	R14	Attacker/calibration transfer matrices, non-oracle write detection and write-only framing with the endpoint suite.	S25
	R15	Fresh paired live-planner episodes under isolated prefix cover and injected tool-response deadlines.	S26

TABLE 23. HOLDOUT EVIDENCE, FAMILIES 1–28. CODES AND AXES ARE DEFINED IN APPENDIX D.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F01 AgentPrint [4]	C	?	?	?	?	?	?	Random trace splits; prompt grouping unresolved. III; V; Appendix D.A–D.D; Fig. 11
F02 UI Traces [5]	C	V	V	V	?	?	?	Cross-task/site comparisons; exact-question overlap remains. 3–4; source index
F03 WRA [6]	C	H	?	?	?	?	?	Prompt tests and synthetic persona split; version reconciliation open. 3.1; 6.1; 8–9; Tables 1/8/10/11; Appendix C.1–C.3
F04 FP-Agent [65]	C	H	?	X	?	?	?	Separate task holdout; visitor/host separation unresolved. 4.1–4.7; Tables 2/4; 6.1–6.2; Appendix E/Table 8
F05 Prompt length [1]	C	H	?	V	?	V	V	OOD inputs; browser/API and day/night buffering controls. 5.1–5.3; Fig. 4; Tables 2–3
F06 PromptPeek [2]	C	V	?	?	?	V	?	Candidate-prior and simulated request-load sweeps. III.B; VI.A–VI.C; Table II
F07 NetEcho [38]	C	V	?	V	?	?	?	Deployment/frontend scenarios; independent split binding unresolved. III.A; Table III; VI.A–VI.B/Table VI
F08 Early Bird [3]	C	?	V	?	?	?	?	Provider observations and candidate calibration; grouped split unresolved. arXiv 2409.20002: II.C; IV.A; V.A; Tables IV–V
F09 InputSnatch [46]	C	H	?	?	?	?	?	Held-out and generated target prompts; candidate-prior dependence. arXiv 2411.18191: 3.2; 4.3–4.4; Tables 2/4
F10 Whisper Leak [41]	C	H	V	?	?	?	?	Unique target-prompt holdout; model/provider comparisons. 3.1; 3 classifier/split discussion; 4.1–4.2; Table 1; 6
F11 Spill the Beans [25]	C	V	?	?	?	?	?	Exposure/monitor sweeps; independent capture grouping unresolved. Threat model; 5/Table 1; 6.1
F12 SpliceLeak [48]	C	V	?	?	?	?	?	Bounded cloze tasks and chunk/candidate coverage. Threat model; VII.A–VII.C; Figs. 7–8
F13 Time Will Tell [49]	C	?	?	?	?	V	V	Concurrent network/load profiling; short-window recalibration. 3; 4.1–4.4; 6.2–6.5; Tables 3–4
F14 Prompt Echo [39]	C	H	V	?	?	?	?	Overlap-filtered prompts; models evaluated separately. 3.1; 5.1–5.3; Tables 2–3
F15 I Know What You Said [22]	C	H	V	V	V	?	?	Prompt overlap filtering; model/hardware/framework comparisons. Abstract; source index
F16 Speculation [40]	C	H	?	?	?	V	?	New traces and OOD paraphrases; remote/load conditions. v4: 4.3–4.8; Table 1; 6.1
F17 Behind Bars [23]	C	?	V	?	V	?	?	Model identification CV; additional hardware evaluation. Table1; 6.1–6.3; Table7 (printed5383/PDF12); 7.1/7.3; Open Science
F18 Prompt caching [45]	C	?	V	?	?	?	V	Provider survey over two months; historical exposure only. 3.1–3.2; 4.1–4.2; Tables 1–2
F19 Whose Agent [63]	C	?	?	?	?	?	?	Trial split; task/site/version independence unresolved. 4,6.1–6.3; Tables4–5; Fig7;7
F20 Workflow traffic [16]	C	?	?	H	?	?	?	Leave-composition-out; stored and live populations separated. 3,9.1,9.7–9.14;Table6
F21 GPU contention [42]	C	?	?	V	?	V	?	Serving-stack and interleaved-load controls. IV,IX,TableVII,X,XIV–XVII
F22 AgentTell [64]	C	V	?	?	?	V	?	Secret rotation; cold/loaded and equal-completability controls. 3–5;AppendixD.2–D.3,Table8;AppendixE
F23 Traffic benchmark [53]	C	?	?	H	?	?	?	Topology shift distinct from feature-view restriction. III,IV-A–C,V-A TablesI–III,V-B Fig8,VI
F24 Remote timing [37]	C	?	V	?	?	?	V	Month-long observations; method/model changes confounded. III,IV-B–C,V-D–E,VI-C,VII,Figs8,10,12
F25 Persona fingerprint [62]	C	?	?	X	?	?	?	Pooled sites; leave-site-out not verified. 2,3.1–3.4,4,5.1,5.5–5.8
F26 TDXRay [29]	C	?	?	?	?	?	?	Tokenizer-localized LLM case; AES result separate. 3;5.1–5.4;6.1–6.2
F27 Detokenization [31]	C	?	?	X	?	?	?	Explicit same-instance profiling restriction. 3.1 and 3.3: shared-code, SMT and same-instance profiling; source index
F28 Spillage [80]	A	V	?	?	?	?	?	Task-irrelevance and selected-site evaluation. 4.1–4.2 (evaluator inputs), 5.4, Table 4, 6.2

TABLE 24. HOLDOUT EVIDENCE, FAMILIES 29–56. CODES AND AXES ARE DEFINED IN APPENDIX D.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F29 PlanTwin [81]	A	V	?	?	?	?	?	Template candidate pool; production transfer unresolved. 2.2,3.2,4.2–4.4,5.1–5.9,6.1
F30 SICC [82]	A	?	?	?	?	?	?	Authorized-view theorem; empirical shift binding unresolved. Theorem 1; Experiments Setup; Tables 3–4, 7–8
F31 TOPR [83]	A	V	?	?	?	?	?	Benchmark and alignment populations differ. 3.1–3.2,4,5.2,5.6,6,Limitations
F32 ClawGuard [15]	C	?	?	V	?	?	V	Carrier recollection and cycle drift; not zero-shot transfer. III-C,IV,VI,AppendixA
F33 SIGLEAK [57]	C	V	?	?	?	?	?	Paired prompts; not physical-skill removal. Definition3,3.2,4.1–4.2
F34 ProcGrep [58]	C	H	?	V	?	?	?	Task-grouped attribution under scaffold regimes. 4–5,Tables7–8
F35 SecEmb [84]	A	?	X	?	?	?	?	Single GPT-2 workload; DLRM result separate. III,V,VI-D;Figs14–15
F36 Mobile agents [85]	A	?	?	?	?	?	?	Direct leakage cases; holdout binding unresolved. IV-D3 log leakage; V-D2 deep-link leakage; V-D3 log leakage; Table II; VI-A
F37 Hidden risks [86]	A	?	?	?	?	?	?	Content exfiltration/access-control cases. 3.2 framework and risk scope; source index
F38 Browser privacy [87]	A	?	?	V	?	?	?	Browser/profile-state measurements; mixed leakage mechanisms. V-C1–3 browser/network/profile-state measurements; source index
F39 LLMs Have Rhythm [36]	C	?	V	?	?	H	V	New captures and cross-network tests; overlapping windows caveat. III-IV; V-C–G; VI-A/B; Table I; Figs10–16; VII
F40 MoEcho [21]	C	H	V	?	?	V	?	Cross-template/dataset tests; contention/context conditions. 3.1;4;6.2–6.6;Tables3,5–10;7.2;8.1–8.2
F41 LLM-Fin [72]	A	?	H	?	?	?	?	Unseen variants within known architecture families. III; IV-B/C; Tables II–III; IV-C/D; V-A; V-A versus Introduction/Fig.2; Table IV
F42 ML privacy channels [44]	C	?	V	?	?	?	?	Enumerative/filter probes across named models; no learned split. 4.1; 4.2.1; 4.2.2/Fig.7; 4.2.3; 4.2.3/Table2
F43 MoE stealing [51]	C	?	?	?	?	?	?	Constructive word extraction in controlled shared batches. 2.2; 4; 4; 4/Table2; 5/Fig.7
F44 Energon [20]	C	?	V	?	?	?	?	Known-candidate CV; exploratory hardware not transfer proof. III-A/B; IV-A/C; IV-B/Tables I,III; V; VII
F45 SafeKV [17]	C	?	V	?	?	?	?	Named serving models; calibration/test separation unreported. 3; 7.1; 7.2.1;7.2.3/Fig.12; 7.3/Table7; 7.4; 4.2.2 versus6/7.2.3
F46 InferNet [88]	A	?	V	?	V	?	?	Same-label mixed-model and platform comparisons. 5.2; source index
F47 ArchSnoop [26]	C	?	H	?	V	?	?	Random-weight training to official/fine-tuned models; power modes. 3.2;6.1; 6.1; Fig9, paperp58, visually checked; 6.2.3;6.3.1; 4.1;7.1-7.2
F48 Coding fingerprints [9]	A	X	?	X	?	?	X	PR CV; repository/time/human-mediated holdouts unreported. 2.1-2.4; Tables 1-2; 4; 6; replication reference
F49 AHB [14]	C	?	?	X	?	?	?	Offline/live humanization; website-transfer binding unresolved. 2.1-2.2; 3.1; 4.1-4.3; 5.1.2; Table 1 Trip Planning; E.1-E.3
F50 WildAuth [10]	A	H	?	X	?	?	?	User-disjoint cross-topic split; no verified cross-service test. 2.3; 4.1-4.3; 5.3-5.5; 6.1-6.2; Tables 1-3; Appendix D/Table4
F51 Cycles [56]	C	X	?	X	?	?	X	One application; no held-out split located. 1; 2; 2.1 CDCS/CDDAG versus CDSA/Hybrid; 3.1; Tables1-2; 3.2; 4
F52 LumiMAS [55]	C	?	V	V	?	?	?	Nonoverlap logs; backend/platform comparisons; task grouping open. 3;4.1-4.3;5.1.1-5.1.7;5.2;Table1;6.1-6.5;B.1/Table3;B.2/Table4;C.1.1
F53 SymDiag [89]	A	V	V	?	?	?	?	Domains/reasoners vary with inputs and feedback. 3.1-3.4;4.1.1-4.1.5;5.1;6.1-6.4;7
F54 LeakyLMs [47]	C	?	H	?	V	?	?	Held-out architecture configurations; known-GPU calibration. 3;4.4-4.5 Tables1-2;5.4-5.5 Tables3-6/Fig6;6;A.1;B.4-B.7
F55 KVGov [35]	C	?	?	?	V	?	?	Simulation, A100 timing and HMAC microcost separate. 2.2-2.3;4;5.1-5.3 Tables3-4;5.7-5.9 Tables8-9;7
F56 Data-centric survey [90]	A	-	-	-	-	-	-	Conceptual synthesis; empirical holdout not applicable. 1 Fig1;4-6;8-9 discussion/conclusions

TABLE 25. HOLDOUT EVIDENCE, FAMILIES 57–82. CODES AND AXES ARE DEFINED IN APPENDIX D.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F57 Serving RCA [33]	C	?	?	X	?	?	?	Single serving deployment with injected faults. Sections3,4.1-4.3,5.1-5.2; Table1
F58 ZeroPrint [11]	A	?	V	?	?	?	?	Known-family provenance pairs; partition not re-extracted. Header;4.1-4.4;Figure1;5.1-5.3
F59 Endpoint fingerprints [12]	A	?	?	V	?	?	V	Controlled endpoint changes over matched time windows. Sections3,4.1-4.3,6.1-6.2/Table1,7-9
F60 StepFinder [13]	A	H	?	?	?	?	?	Task-disjoint subsets; denominator reconciliation unresolved. Header;Sections4,5.1-5.4,6;Tables4-5
F61 Web-agent detection [54]	C	?	V	?	?	?	?	Six agents with human/scraper controls. 3.1.4;3.3;6;7
F62 Cross-campaign [61]	C	V	?	V	?	?	?	Crossed-style and regenerated conditions; execution stress subset. AppendixB decomposition; source index
F63 AgentProv [59]	C	?	V	?	?	?	?	Substituted checkpoints; system-prompt perturbation. 3.1-3.4;4.1-4.4;A.5
F64 LiDAR [60]	C	?	V	V	?	?	?	Enrollment-only identity; two harnesses and perturbations. 2;5.1-5.3;6.1-6.5
F65 LLM-WFIN [32]	C	V	?	?	?	?	?	Site traces; loading versus query phases. 3.2-3.3;5.1-5.2;6.1-6.5;Tables2-5;publisher rendered primary article
F66 MIRAGE [52]	C	V	V	?	?	?	?	Encodings/models/scenarios; matched decoder-layer controls. 3.4;4.1-4.2;7.1-7.4;AppendixL.3-L.4
F67 Routing metadata [43]	C	V	?	V	?	?	?	Request domains and routers; exploratory sensitive-label gate. III;IV;V;VI;VII
F68 Shape and Substance [28]	C	V	V	V	V	?	?	Geometry/model/cache/hardware conditions; padding estimate. IV;V;VII;VIII;IX
F69 SplitDecoy [34]	C	?	?	?	?	?	?	Repeated seeds; forward/backward boundary, not a holdout. 2;5.1-5.3;Table7-8;A.2.2
F70 PAGE404 [24]	C	V	V	?	?	?	?	Architecture and prompt-length targets; transfer binding open. II-B;IV-D4-D6;V
F71 Edge EM [19]	C	H	?	?	?	?	?	Unseen-word test separate from profiled vocabulary recovery. IV;V;VI
F72 SparLeak [27]	C	V	V	V	?	?	?	Models/datasets/sparse mechanisms; per-phase outcomes. 3.1;4-7;Table4
F73 Sparseety [30]	C	V	V	V	?	?	?	Models/datasets/implementations; oracle-conditioned extraction. III;V;VI;VIII;IX
F74 OTRO [18]	C	?	?	?	?	?	?	Closed-corpus conditional leakage; epoch cost separately derived. III;V;VI;VII-C-E;TableIII
F75 CachePrune [50]	C	V	V	?	?	?	?	Models/workloads and detector-miss stress; separate fits. 3.1;5.1-5.6;Table1;AppendixA
F76 Privacy foundations [68]	A	-	-	-	-	-	-	Conceptual contract; cross-institution example illustrative. III;IV/TableIII;IV-A;VI
F77 DP-MemView [91]	A	?	?	?	?	?	?	Semantic response privacy; holdout binding not re-extracted. Problem Definition;DP-MemView;Experiments;AppendixB.5
F78 PriCon [92]	A	?	?	?	?	?	?	Semantic cloud consultation; holdout binding not re-extracted. PDF3;5.1-5.2;Tables2-4
F79 SCAAR [93]	A	?	?	?	?	?	?	LLM EM/length test; other classifier results separate. 7;LLM extension;TVLA experiment
F80 CacheSolidarity [94]	A	?	?	?	?	?	?	Candidate-prefix timing gap; no scored inference error. v1 title;5.2.3;Figure7
F81 Vision encoder [95]	A	?	?	?	?	?	?	Representation inversion; holdout binding not re-extracted. 2;3;5;7;8/Table5
F82 Split gradients [96]	A	?	?	?	?	?	?	Split-compute content inversion; holdout binding unresolved. 3;4.1-4.5;5;AppendixB